Cyclone V SoC Development & Education Board (DE0-Nano-SoC)

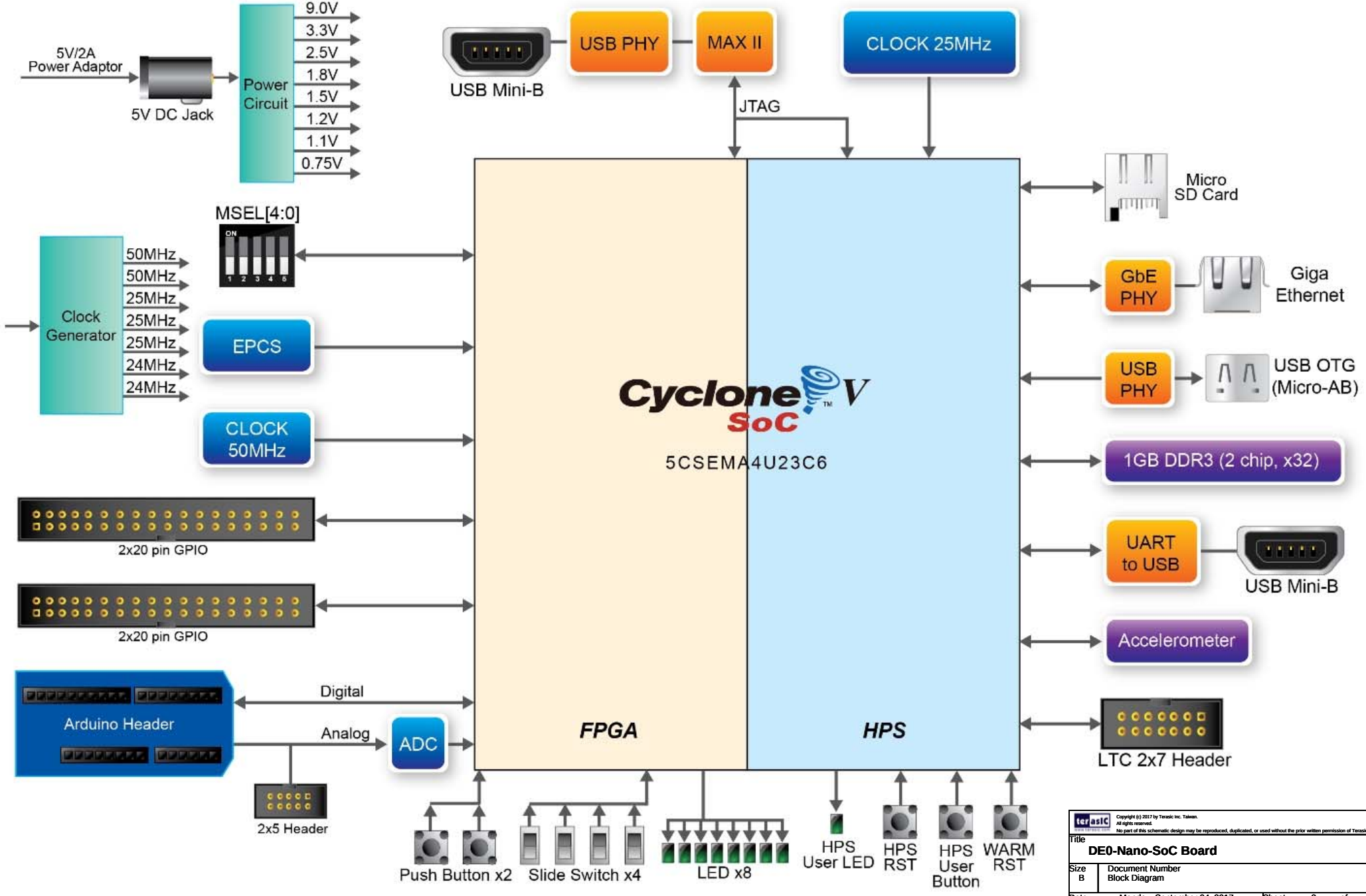
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17	HPS Peripheral : Reset Circuit, Button and LED
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22	Power - 2.5V, 3.3V
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24	

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Title: **DE0-Nano-SoC Board**

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Title
DE0-Nano-SoC Board

Size	Document Number	Rev
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CYCLONE V SoC BANK 3

Bank 3A
VCCIO = 3.3V

Y11	IO_3A/PR_ERROR/DIFFIO_RX_B7P	
AA11	IO_3A/PR_DONE/DIFFIO_RX_B7N	
AD5	IO_3A/DIFFIO_TX_B8P/DQ1B	
AE6	IO_3A/PR_READY/DIFFIO_TX_B8N/DQ1B	

Bank 3B
VCCIO = 3.3V

GPIO_0_D21	AE4	IO_3B/DIFFIO_TX_B9P/B_WEN/DQ2B	IO_3B/DIFFIO_TX_B17P/B_BA_0/DQ3B	AF7 GPIO_0_D1
GPIO_0_D18	AF4	IO_3B/DIFFIO_TX_B9N/GND	IO_3B/DIFFIO_TX_B17N/GND	AG6 GPIO_0_D19
GPIO_0_D28	AD10	IO_3B/DIFFIO_RX_B10P/B_A_14/DQ2B	IO_3B/DIFFIO_RX_B18P/B_BA_1/DQ3B	AF11 GPIO_0_D34
GPIO_0_D29	AE9	IO_3B/DIFFIO_RX_B10N/B_A_15/DQ2B	IO_3B/DIFFIO_RX_B18N/B_BA_2/DQ3B	AF10 GPIO_0_D31
GPIO_0_D23	T11	IO_3B/DIFFIO_RX_B11P/B_CSN_0/DQS2B	IO_3B/DIFFIO_RX_B19P/B_CK/DQS3B	T13 GPIO_0_D22
GPIO_0_D9	U11	IO_3B/DIFFIO_RX_B11N/B_CSN_1/DQSN2B	IO_3B/DIFFIO_RX_B19N/B_CKN/DQSN3B	T12 GPIO_0_D11
GPIO_0_D24	AE7	IO_3B/DIFFIO_TX_B12P/B_A_12	IO_3B/DIFFIO_TX_B20P/B_A_6	AH3 GPIO_0_D16
GPIO_0_D3	AF8	IO_3B/DIFFIO_TX_B12N/B_A_13/DQ2B	IO_3B/DIFFIO_TX_B20N/B_A_7/DQ3B	AH2 GPIO_0_D17
GPIO_0_D27	AE8	IO_3B/DIFFIO_TX_B13P/B_A_10/DQ2B		
GPIO_0_D26	AF9	IO_3B/DIFFIO_TX_B13N/B_A_11/DQ2B		
GPIO_0_D30	AD11	IO_3B/DIFFIO_RX_B14P/B_A_8/DQ2B	IO_3B/DIFFIO_RX_B22P/B_A_4/DQ3B	AE12 GPIO_0_D35
GPIO_0_D33	AE11	IO_3B/DIFFIO_RX_B14N/B_A_9/DQ2B	IO_3B/DIFFIO_RX_B22N/B_A_5/DQ3B	AD12 GPIO_0_D32
GPIO_0_D20	AF5	IO_3B/DIFFIO_TX_B16P/B_CASN/DQ2B	IO_3B/DIFFIO_TX_B24P/B_A_0/DQ3B	AH6 GPIO_0_D13
GPIO_0_D25	AF6	IO_3B/DIFFIO_TX_B16N/B_RASN/DQ2B	IO_3B/DIFFIO_TX_B24N/B_A_1/DQ3B	AH5 GPIO_0_D12

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CYCLONE V SoC BANK 5

Bank 5A
VCCIO = 3.3V

LED4	AF26	IO_5A/RZQ_1/DIFFIO_TX_R1P/DQ1R		
LED5	AE26	IO_5A/PR_REQUEST/DIFFIO_TX_R1N/DQ1R		
			IO_5A/DIFFIO_RX_R6P/DQS1R	Y16 LED6
			IO_5A/DIFFIO_RX_R6N/DQSN1R	W15 LED0
			IO_5A/DIFFIO_TX_R7P/DQ1R	AA24 LED1
	AD26			AA23 LED7
	Y17	IO_5A/CVP_CONFDONE/DIFFIO_TX_R3N/DQ1R	IO_5A/DIFFIO_TX_R7N	V16 LED2
	Y18	IO_5A/DIFFIO_RX_R4P/DQ1R	IO_5A/DIFFIO_RX_R8P/DQ1R	V15 LED3
		IO_5A/DIFFIO_RX_R4N/DQ1R	IO_5A/DIFFIO_RX_R8N/DQ1R	

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User Interface (FPGA)

20	KEY[1..0]
20	SW[3..0]
20	LED[7..0]

GPIO

6,7,19	GPIO_0_D[35..0]
6,19	GPIO_1_D[35..0]

Arduino Digital Interface

19	Arduino_IO[15..0]
19	Arduino_Reset_n

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CYCLONE V SoC BANK 4

Bank 4A
VCCIO = 3.3V

Arduino_Reset_n	AH7	IO_4A/RZQ_0/DIFFIO_TX_B25N		
Arduino_IO6	AG8	IO_4A/DIFFIO_TX_B25P/B_DQ_2/DQ4B	IO_4A/DIFFIO_TX_B41P/B_DQ_18/DQ6B	AF18 GPIO_1_D21
Arduino_IO0	AG13			AE19 GPIO_1_D20
Arduino_IO1	AF13	IO_4A/DIFFIO_RX_B26P/B_DQ_1/DQ4B	IO_4A/DIFFIO_RX_B42P/B_DQ_17/DQ6B	AD19 GPIO_1_D22
Arduino_IO4	U14	IO_4A/DIFFIO_RX_B26N/B_DQ_0/DQ4B	IO_4A/DIFFIO_RX_B42N/B_DQ_16/DQ6B	AA19 GPIO_1_D15
Arduino_IO5	U13	IO_4A/DIFFIO_RX_B27P/B_DQS_0/DQS4B	IO_4A/DIFFIO_RX_B43P/B_DQS_2/DQS6B	AA18 GPIO_1_D34
Arduino_IO3	AG9	IO_4A/DIFFIO_RX_B27N/B_DQSN_0/DQSN4B	IO_4A/DIFFIO_RX_B43N/B_DQSN_2/DQSN6B	AG18 GPIO_1_D13
Arduino_IO7	AH8	IO_4A/DIFFIO_TX_B28P/B_ODT_0	IO_4A/DIFFIO_TX_B44P/B_RESETN	AH18 GPIO_1_D31
Arduino_IO2	AG10	IO_4A/DIFFIO_TX_B28N/B_DQ_3/DQ4B	IO_4A/DIFFIO_TX_B44N/B_DQ_19/DQ6B	AG19 GPIO_1_D10
Arduino_IO14	AH9	IO_4A/DIFFIO_TX_B29P/B_DQ_6/DQ4B	IO_4A/DIFFIO_TX_B45P/B_DQ_22/DQ6B	AH19 GPIO_1_D28
Arduino_IO10	AF15	IO_4A/DIFFIO_TX_B29N/B_ODT_1/DQ4B	IO_4A/DIFFIO_TX_B45N/GND/DQ6B	AE20 GPIO_1_D23
Arduino_IO9	AE15	IO_4A/DIFFIO_RX_B30P/B_DQ_5/DQ4B	IO_4A/DIFFIO_RX_B46P/B_DQ_21/DQ6B	AD20 GPIO_1_D25
		IO_4A/DIFFIO_RX_B30N/B_DQ_4/DQ4B	IO_4A/DIFFIO_RX_B46N/B_DQ_20/DQ6B	
Arduino_IO15	AG11			AF20 GPIO_1_D11
Arduino_IO12	AH11	IO_4A/DIFFIO_TX_B32P/B_DM_0/DQ4B	IO_4A/DIFFIO_TX_B48P/B_DM_2/DQ6B	AG20 GPIO_1_D9
		IO_4A/DIFFIO_TX_B32N/B_DQ_7/DQ4B	IO_4A/DIFFIO_TX_B48N/B_DQ_23/DQ6B	
Arduino_IO13	AH12			AG21 GPIO_1_D30
		IO_4A/DIFFIO_TX_B33P/B_DQ_10/DQ5B	IO_4A/DIFFIO_TX_B49P/B_DQ_26/DQ7B	
Arduino_IO8	AF17			AF22 GPIO_1_D26
Arduino_IO11	AG16	IO_4A/DIFFIO_RX_B34P/B_DQ_9/DQ5B	IO_4A/DIFFIO_RX_B50P/B_DQ_25/DQ7B	AF21 GPIO_1_D8
	W14	IO_4A/DIFFIO_RX_B34N/B_DQ_8/DQ5B	IO_4A/DIFFIO_RX_B50N/B_DQ_24/DQ7B	AD23 GPIO_1_D32
	V13	IO_4A/DIFFIO_RX_B35P/B_DQS_1/DQS5B	IO_4A/DIFFIO_RX_B51P/B_DQS_3/DQS7B	AE22 GPIO_1_D7
	AG14	IO_4A/DIFFIO_RX_B35N/B_DQSN_1/DQSN5B	IO_4A/DIFFIO_RX_B51N/B_DQSN_3/DQSN7B	
	AH13	IO_4A/DIFFIO_TX_B36P/B_CKE_1		AH21 GPIO_1_D29
	AG15	IO_4A/DIFFIO_TX_B36N/B_DQ_11/DQ5B	IO_4A/DIFFIO_TX_B52N/B_DQ_27/DQ7B	AH23 GPIO_1_D18
	AH14	IO_4A/DIFFIO_TX_B37P/B_DQ_14/DQ5B	IO_4A/DIFFIO_TX_B53P/B_DQ_30/DQ7B	AH22 GPIO_1_D27
	AD17	IO_4A/DIFFIO_TX_B37N/B_CKE_0/DQ5B	IO_4A/DIFFIO_TX_B53N/GND/DQ7B	AG23 GPIO_1_D19
	AE17	IO_4A/DIFFIO_RX_B38P/B_DQ_13/DQ5B	IO_4A/DIFFIO_RX_B54P/B_DQ_29/DQ7B	AF23 GPIO_1_D6
		IO_4A/DIFFIO_RX_B38N/B_DQ_12/DQ5B	IO_4A/DIFFIO_RX_B54N/B_DQ_28/DQ7B	
KEY0	AH17			AG24 GPIO_1_D16
KEY1	AH16	IO_4A/DIFFIO_TX_B40P/B_DM_1/DQ5B	IO_4A/DIFFIO_TX_B56P/B_DM_3/DQ7B	AH24 GPIO_1_D5
		IO_4A/DIFFIO_TX_B40N/B_DQ_15/DQ5B	IO_4A/DIFFIO_TX_B56N/B_DQ_31/DQ7B	
				AG26 GPIO_1_D4
			IO_4A/DIFFIO_TX_B57P/B_DQ_34/DQ8B	
				AE24 GPIO_1_D24
			IO_4A/DIFFIO_RX_B58P/B_DQ_33/DQ8B	AE23 GPIO_1_D33
			IO_4A/DIFFIO_RX_B58N/B_DQ_32/DQ8B	AC22 GPIO_1_D35
			IO_4A/DIFFIO_RX_B59P/B_DQS_4/DQS8B	AC23 GPIO_1_D12
			IO_4A/DIFFIO_RX_B59N/B_DQSN_4/DQSN8B	
				AH26 GPIO_1_D14
			IO_4A/DIFFIO_TX_B60N/B_DQ_35/DQ8B	AG28 GPIO_1_D1
			IO_4A/DIFFIO_TX_B61P/B_DQ_38/DQ8B	AH27 GPIO_1_D3
			IO_4A/DIFFIO_TX_B61N/GND/DQ8B	AF25 GPIO_1_D17
			IO_4A/DIFFIO_RX_B62P/B_DQ_37/DQ8B	AG25
			IO_4A/DIFFIO_RX_B62N/B_DQ_36/DQ8B	
				AF27
			IO_4A/DIFFIO_TX_B64P/B_DM_4/DQ8B	AF28
			IO_4A/DIFFIO_TX_B64N/B_DQ_39/DQ8B	

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CYCLONE V SoC BANK 8

Bank 8A
VCCIO = 3.3V

SW0	L10	IO_8A/DIFFIO_RX_T21P	IO_8A/DIFFIO_RX_T23P	L8
SW1	L9	IO_8A/DIFFIO_RX_T21N	IO_8A/DIFFIO_RX_T23N	K8
SW2	H6	IO_8A/DIFFIO_TX_T22P		
SW3	H5	IO_8A/DIFFIO_TX_T22N	IO_8A/DIFFIO_TX_T24N	H4

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Title: **DE0-Nano-SoC Board**

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B	FPGA Bank 3, 4, 5, 8	A2

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DDR3 Interface (HPS)

12	HPS_DDR3_ADDR[14..0]
12	HPS_DDR3_BA[2..0]
12	HPS_DDR3_DM[3..0]
12	HPS_DDR3_DQ[31..0]
12	HPS_DDR3_DQS_P[3..0]
12	HPS_DDR3_DQS_N[3..0]
12	HPS_DDR3_RESET_N
12	HPS_DDR3_CK_P
12	HPS_DDR3_CK_N
12	HPS_DDR3_CAS_N
12	HPS_DDR3_RAS_N
12	HPS_DDR3_CS_N
12	HPS_DDR3_WE_N
12	HPS_DDR3_ODT
12	HPS_DDR3_CKE
12	HPS_DDR3_CS1_N
12	HPS_DDR3_CKE1
12	HPS_DDR3_ODT1

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CYCLONE V SoC BANK 6 (HPS)

R117 100 HPS_DDR3_RZQ D25 — HPS_RZQ_0

Bank 6A
VCCIO = 1.5V

HPS_DDR3_ADDR0	C28	HPS_DDR/HPS_A_0	HPS_DDR/HPS_DQ_0	J25	HPS_DDR3_DQ0
HPS_DDR3_ADDR1	B28	HPS_DDR/HPS_A_1	HPS_DDR/HPS_DQ_1	J24	HPS_DDR3_DQ1
HPS_DDR3_ADDR2	E26	HPS_DDR/HPS_A_2	HPS_DDR/HPS_DQ_2	E28	HPS_DDR3_DQ2
HPS_DDR3_ADDR3	D26	HPS_DDR/HPS_A_3	HPS_DDR/HPS_DQ_3	D27	HPS_DDR3_DQ3
HPS_DDR3_ADDR4	J21	HPS_DDR/HPS_A_4	HPS_DDR/HPS_DQ_4	J26	HPS_DDR3_DQ4
HPS_DDR3_ADDR5	J20	HPS_DDR/HPS_A_5	HPS_DDR/HPS_DQ_5	K26	HPS_DDR3_DQ5
HPS_DDR3_ADDR6	C26	HPS_DDR/HPS_A_6	HPS_DDR/HPS_DQ_6	G27	HPS_DDR3_DQ6
HPS_DDR3_ADDR7	B26	HPS_DDR/HPS_A_7	HPS_DDR/HPS_DQ_7	F28	HPS_DDR3_DQ7
HPS_DDR3_ADDR8	F26	HPS_DDR/HPS_A_8			
HPS_DDR3_ADDR9	F25	HPS_DDR/HPS_A_9	HPS_DDR/HPS_DQS_0	R17	HPS_DDR3_DQS_P0
HPS_DDR3_ADDR10	A24	HPS_DDR/HPS_A_10	HPS_DDR/HPS_DQSn_0	R16	HPS_DDR3_DQS_N0
HPS_DDR3_ADDR11	B24	HPS_DDR/HPS_A_11	HPS_DDR/HPS_DM_0	G28	HPS_DDR3_DM0
HPS_DDR3_ADDR12	D24	HPS_DDR/HPS_A_12			
HPS_DDR3_ADDR13	C24	HPS_DDR/HPS_A_13			
HPS_DDR3_ADDR14	G23	HPS_DDR/HPS_A_14	HPS_DDR/HPS_DQ_8	K25	HPS_DDR3_DQ8
	F24	HPS_DDR/HPS_A_15	HPS_DDR/HPS_DQ_9	L25	HPS_DDR3_DQ9
			HPS_DDR/HPS_DQ_10	J27	HPS_DDR3_DQ10
HPS_DDR3_CK_P	N21	HPS_DDR/HPS_CK	HPS_DDR/HPS_DQ_11	J28	HPS_DDR3_DQ11
HPS_DDR3_CK_N	N20	HPS_DDR/HPS_CKn	HPS_DDR/HPS_DQ_12	M27	HPS_DDR3_DQ12
HPS_DDR3_CKE	L28	HPS_DDR/HPS_CKE_0	HPS_DDR/HPS_DQ_13	M26	HPS_DDR3_DQ13
HPS_DDR3_CKE1	K28	HPS_DDR/HPS_CKE_1	HPS_DDR/HPS_DQ_14	M28	HPS_DDR3_DQ14
HPS_DDR3_BA0	A27	HPS_DDR/HPS_BA_0	HPS_DDR/HPS_DQ_15	N28	HPS_DDR3_DQ15
HPS_DDR3_BA1	H25	HPS_DDR/HPS_BA_1			
HPS_DDR3_BA2	G25	HPS_DDR/HPS_BA_2	HPS_DDR/HPS_DQS_1	R19	HPS_DDR3_DQS_P1
HPS_DDR3_RAS_N	A25	HPS_DDR/HPS_RASn	HPS_DDR/HPS_DQSn_1	R18	HPS_DDR3_DQS_N1
HPS_DDR3_CAS_N	A26	HPS_DDR/HPS_CASn	HPS_DDR/HPS_DM_1	P28	HPS_DDR3_DM1
HPS_DDR3_WE_N	E25	HPS_DDR/HPS_WEn			
HPS_DDR3_CS_N	L21	HPS_DDR/HPS_CSn_0	HPS_GPI10	U15	
HPS_DDR3_CS1_N	L20	HPS_DDR/HPS_CSn_1	HPS_GPI11	U16	
HPS_DDR3_ODT	D28	HPS_DDR/HPS_ODT_0	HPS_GPI12	AC27	
HPS_DDR3_ODT1	G26	HPS_DDR/HPS_ODT_1	HPS_GPI13	V24	

Bank 6B
VCCIO = 1.5V

HPS_DDR3_DQ16	N24	HPS_DDR/HPS_DQ_16	HPS_DDR/HPS_DQ_32	T26	
HPS_DDR3_DQ17	N25	HPS_DDR/HPS_DQ_17	HPS_DDR/HPS_DQ_33	U25	
HPS_DDR3_DQ18	T28	HPS_DDR/HPS_DQ_18	HPS_DDR/HPS_DQ_34	AC28	
HPS_DDR3_DQ19	U28	HPS_DDR/HPS_DQ_19	HPS_DDR/HPS_DQ_35	V25	
HPS_DDR3_DQ20	N26	HPS_DDR/HPS_DQ_20	HPS_DDR/HPS_DQ_36	V19	
HPS_DDR3_DQ21	N27	HPS_DDR/HPS_DQ_21	HPS_DDR/HPS_DQ_37	V20	
HPS_DDR3_DQ22	R27	HPS_DDR/HPS_DQ_22	HPS_DDR/HPS_DQ_38	AE27	
HPS_DDR3_DQ23	V27	HPS_DDR/HPS_DQ_23	HPS_DDR/HPS_DQ_39	AD28	
HPS_DDR3_DQS_P2	T19	HPS_DDR/HPS_DQS_2	HPS_DDR/HPS_DQS_4	V18	
HPS_DDR3_DQS_N2	T18	HPS_DDR/HPS_DQSn_2	HPS_DDR/HPS_DQSn_4	V17	
HPS_DDR3_DM2	W28	HPS_DDR/HPS_DM_2	HPS_DDR/HPS_DM_4	AE28	
HPS_DDR3_DQ24	R26	HPS_DDR/HPS_DQ_24	HPS_GPI0	M25	
HPS_DDR3_DQ25	R25	HPS_DDR/HPS_DQ_25	HPS_GPI1	K27	
HPS_DDR3_DQ26	AA28	HPS_DDR/HPS_DQ_26	HPS_GPI2	R20	
HPS_DDR3_DQ27	W26	HPS_DDR/HPS_DQ_27	HPS_GPI3	R21	
HPS_DDR3_DQ28	R24	HPS_DDR/HPS_DQ_28	HPS_GPI4	R28	
HPS_DDR3_DQ29	T24	HPS_DDR/HPS_DQ_29	HPS_GPI5	P26	
HPS_DDR3_DQ30	Y27	HPS_DDR/HPS_DQ_30	HPS_GPI6	T17	
HPS_DDR3_DQ31	AA27	HPS_DDR/HPS_DQ_31	HPS_GPI7	T16	
			HPS_GPI8	Y28	
HPS_DDR3_DQS_P3	U19	HPS_DDR/HPS_DQS_3	HPS_GPI9	Y26	
HPS_DDR3_DQS_N3	T20	HPS_DDR/HPS_DQSn_3			
HPS_DDR3_DM3	AB28	HPS_DDR/HPS_DM_3			
			HPS_DDR/HPS_RESETn	V28	HPS_DDR3_RESET_N

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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	FPGA Bank 6	A2

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UBS PHY Interface (ULPI)

14 HPS_USB_DATA[7..0]
14 HPS_USB_CLKOUT
14 HPS_USB_NXT
14 HPS_USB_DIR
14 HPS_USB_STP
10,14 HPS_USB_RESET

Ethernet PHY Interface (RGMII)

15 HPS_ENET_TX_DATA[3..0]
15 HPS_ENET_GTX_CLK
15 HPS_ENET_TX_EN
15 HPS_ENET_RX_DATA[3..0]
15 HPS_ENET_RX_CLK
15 HPS_ENET_RX_DV
15 HPS_ENET_INT_N
15 HPS_ENET_MDC
15 HPS_ENET_MDIO
10,15,17 HPS_ENET_RESET_N

SD Card Interface

13 HPS_SD_DATA[3..0]
13 HPS_SD_CMD
13 HPS_SD_CLK

UART Interface

13 HPS_UART_RX
13 HPS_UART_TX
13 HPS_CONV_USB_N

HPS Reset

10,17 HPS_WARM_RST_N
10,17 HPS_RESET_N

HPS Clock

6 HPS_CLK1_25
6 HPS_CLK2_25

HPS JTAG INTERFACE

11 HPS_TCK
11 HPS_TMS
11 HPS_TDI
11 HPS_TDO
10 JTAG_TRST

Accelerometer Interface

16 HPS_I2C0_SDAT
16 HPS_I2C0_SCLK
16 HPS_GSENSOR_INT

LTC Interface

16 HPS_I2C1_SDAT
16 HPS_I2C1_SCLK
16 HPS_SPIM_MOSI
16 HPS_SPIM_MISO
16 HPS_SPIM_CLK
16 HPS_SPIM_SS
16 HPS_LTC_GPIO

HPS Key and LED

17 HPS_KEY
17 HPS_LED

VCC3P3
R143 4.7K HPS_PORSEL
R118 4.7K JTAG_TRST
R119 0 DNI HPS_TCK HPS_TMS HPS_TDI HPS_TDO
C89 33p 50V DNI

U1M

CYCLONE V SoC BANK 7 (HPS)

Bank 7A
VCCIO = 3.3V

HPS_WARM_RST_N A23
HPS_RESET_N H19
HPS_CLK1_25 E20
HPS_CLK2_25 D20
HPS_PORSEL E18
JTAG_TRST C22
HPS_TCK K19
HPS_TMS C23
HPS_TDI D22
HPS_TDO B23

HPS_NRST
HPS_NPOR
HPS_CLK1
HPS_CLK2
HPS_PORSEL
HPS_TRST
HPS_TCK
HPS_TMS
HPS_TDI
HPS_TDO

TRACE_CLK/ - / - /HPS_GPIO48 C21
TRACE_D0/SPIS0_CLK/UART0_RX/HPS_GPIO49 A22 HPS_UART_RX
TRACE_D1/SPIS0_MOSI/UART0_TX/HPS_GPIO50 B21 HPS_UART_TX
TRACE_D2/SPIS0_MISO/I2C1_SDA/HPS_GPIO51 A21 HPS_I2C1_SDAT
TRACE_D3/SPIS0_SS0/I2C1_SCL/HPS_GPIO52 K18 HPS_I2C1_SCLK
TRACE_D4/SPIS1_CLK/CAN1_RX/HPS_GPIO53 A20 HPS_LED
TRACE_D5/SPIS1_MOSI/CAN1_TX/HPS_GPIO54 J18 HPS_KEY
TRACE_D6/SPIS1_SS0/I2C0_SDA/HPS_GPIO55 A19 HPS_I2C0_SDAT
TRACE_D7/SPIS1_MISO/I2C0_SCL/HPS_GPIO56 C18 HPS_I2C0_SCLK
SPIM0_CLK/I2C1_SDA/UART0_CTS/HPS_GPIO57 A18
SPIM0_MOSI/I2C1_SCL/UART0_RTS/HPS_GPIO58 C17
SPIM0_MISO/CAN1_RX/UART1_CTS/HPS_GPIO59 B18
SPIM0_SS0,BOOTSEL0/CAN1_TX/UART1_RTS/HPS_GPIO60 J17 HPS_BOOTSEL0
UART0_RX/CAN0_RX/SPIM0_SS1/HPS_GPIO61 A17 HPS_GSENSOR_INT
UART0_TX,CLKSEL1/CAN0_TX/SPIM1_SS1/HPS_GPIO62 H17 HPS_CLOCKSEL1
I2C0_SDA/UART1_RX/SPIM1_CLK/HPS_GPIO63 C19 HPS_SPIM_CLK
I2C0_SCL/UART1_TX/SPIM1_MOSI/HPS_GPIO64 B16 HPS_SPIM_MOSI
CAN0_RX/UART0_RX/SPIM1_MISO/HPS_GPIO65 B19 HPS_SPIM_MISO
CAN0_TX,CLKSEL0/UART0_TX/SPIM1_SS0/HPS_GPIO66 C16 HPS_SPIM_SS

Bank 7B
VCCIO = 3.3V

HPS_ENET_GTX_CLK J15
HPS_ENET_TX_DATA0 A16
HPS_ENET_TX_DATA1 J14
HPS_ENET_TX_DATA2 A15
HPS_ENET_TX_DATA3 D17
HPS_ENET_RX_DATA0 A14
HPS_ENET_MDIO E16
HPS_ENET_MDC A13
HPS_ENET_RX_DV J13
HPS_ENET_TX_EN A12
HPS_ENET_RX_CLK J12
HPS_ENET_RX_DATA1 A11
HPS_ENET_RX_DATA2 C15
HPS_ENET_RX_DATA3 A9
HPS_BOOTSEL2 D15

NAND_ALE/RGMII1_TX_CLK/QSPI_SS3/HPS_GPIO14
NAND_CE/RGMII1_TXD0/USB1_D0/HPS_GPIO15
NAND_CLE/RGMII1_TXD1/USB1_D1/HPS_GPIO16
NAND_RE/RGMII1_TXD2/USB1_D2/HPS_GPIO17
NAND_RB/RGMII1_TXD3/USB1_D3/HPS_GPIO18
NAND_DQ0/RGMII1_RXD0/ - /HPS_GPIO19
NAND_DQ1/RGMII1_MDIO/I2C3_SDA/HPS_GPIO20
NAND_DQ2/RGMII1_MDC/I2C3_SCL/HPS_GPIO21
NAND_DQ3/RGMII1_RX_CTL/USB1_D4/HPS_GPIO22
NAND_DQ4/RGMII1_TX_CTL/USB1_D5/HPS_GPIO23
NAND_DQ5/RGMII1_RX_CLK/USB1_D6/HPS_GPIO24
NAND_DQ6/RGMII1_RXD1/USB1_D7/HPS_GPIO25
NAND_DQ7/RGMII1_RXD2/ - /HPS_GPIO26
NAND_WP/RGMII1_RXD3/QSPI_SS2/HPS_GPIO27
NAND_WE,BOOTSEL2/QSPI_SS1/ - /HPS_GPIO28

QSPI_IO0/ - /USB1_CLK/HPS_GPIO29 A8
QSPI_IO1/ - /USB1_STP/HPS_GPIO30 H16
QSPI_IO2/ - /USB1_DIR/HPS_GPIO31 A7
QSPI_IO3/ - /USB1_NXT/HPS_GPIO32 J16
QSPI_SS0,BOOTSEL1/ - / - /HPS_GPIO33 A6 HPS_BOOTSEL1
QSPI_CLK/ - / - /HPS_GPIO34 C14
QSPI_SS1/ - / - /HPS_GPIO35 B14 HPS_ENET_INT_N

Bank 7C
VCCIO = 3.3V

HPS_SD_CMD D14
HPS_SD_DATA0 C13
HPS_SD_DATA1 B6
HPS_SD_DATA2 B11
HPS_SD_DATA3 B9
B12

SDMMC_CMD/USB0_D0/ - /HPS_GPIO36
SDMMC_D0/USB0_D2/ - /HPS_GPIO38
SDMMC_D1/USB0_D3/ - /HPS_GPIO39
SDMMC_D2/USB0_DIR/ - /HPS_GPIO46
SDMMC_D3/USB0_NXT/ - /HPS_GPIO47
SDMMC_FB_CLK_IN/USB0_CLK/ - /HPS_GPIO44

SDMMC_PWREN/USB0_D1/ - /HPS_GPIO37 A5
SDMMC_D4/USB0_D4/ - /HPS_GPIO40 H13 HPS_LTC_GPIO
SDMMC_D5/USB0_D5/ - /HPS_GPIO41 A4
SDMMC_D6/USB0_D6/ - /HPS_GPIO42 H12 R120 0 DNI HPS_USB_RESET
SDMMC_D7/USB0_D7/ - /HPS_GPIO43 B4 R121 0 DNI HPS_ENET_RESET_N
SDMMC_CCLK_OUT/USB0_STP/ - /HPS_GPIO45 B8 HPS_SD_CLK

Bank 7D
VCCIO = 3.3V

HPS_USB_DATA4 C8
HPS_USB_STP C5
HPS_USB_DIR E5
HPS_USB_NXT D5
HPS_USB_CLKOUT G4
HPS_USB_DATA7 F4
HPS_USB_DATA5 D4
HPS_USB_DATA6 C7

RGMII0_RXD0/USB1_D4/ - /HPS_GPIO5
RGMII0_RXD1/USB1_STP/ - /HPS_GPIO11
RGMII0_RXD2/USB1_DIR/ - /HPS_GPIO12
RGMII0_RXD3/USB1_NXT/ - /HPS_GPIO13
RGMII0_RX_CLK/USB1_CLK/ - /HPS_GPIO10
RGMII0_RX_CTL/USB1_D7/ - /HPS_GPIO8
RGMII0_MDIO/USB1_D5/I2C2_SDA/HPS_GPIO6
RGMII0_MDC/USB1_D6/I2C2_SCL/HPS_GPIO7

RGMII0_TXD0/USB1_D0/ - /HPS_GPIO1 C10 HPS_USB_DATA0
RGMII0_TXD1/USB1_D1/ - /HPS_GPIO2 F5 HPS_USB_DATA1
RGMII0_TXD2/USB1_D2/ - /HPS_GPIO3 C9 HPS_USB_DATA2
RGMII0_TXD3/USB1_D3/ - /HPS_GPIO4 C4 HPS_USB_DATA3
RGMII0_TX_CTL/ - / - /HPS_GPIO9 C6 HPS_CONV_USB_N
RGMII0_TX_CLK/ - / - /HPS_GPIO0 E4

5CSEMA4U23C6N

Default Setting: BOOTSEL[2:0]=101 (Boot from SD CARD)
CLKSEL[1:0] =00

VCC3P3
DNI R127 1K HPS_BOOTSEL0 (BOOTSEL0) R140 10K
R126 1K HPS_BOOTSEL1 (BOOTSEL1) R139 10K DNI
DNI R125 1K HPS_BOOTSEL2 (BOOTSEL2) R138 10K
R124 1K HPS_SPIM_SS (CLOCKSEL0) R137 10K DNI
R123 1K HPS_CLOCKSEL1 (CLOCKSEL1) R135 10K DNI

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Title: DE0-Nano-SoC Board
Size: B | Document Number: FPGA Bank 7 | Rev: A2
Date: Monday, September 04, 2017 | Sheet 5 of 23

GPIO

3,7,19 GPIO_0_D[35..0]

3,19 GPIO_1_D[35..0]

U1B

CYCLONE V SoC Clock

Bank 3B

FPGA_CLK1_50	V11	IO_3B/CLK0P,FPLL_BL_FBP/DIFFIO_RX_B15P	IO_3B/FPLL_BL_CLKOUT0,FPLL_BL_CLKOUTP,FPLL_BL_FB/DIFFIO_TX_B21P/B_A_2/DQ3B	AG5	GPIO_0_D15
	W11	IO_3B/CLK0N,FPLL_BL_FBN/DIFFIO_RX_B15N	IO_3B/FPLL_BL_CLKOUT1,FPLL_BL_CLKOUTN/DIFFIO_TX_B21N/B_A_3/DQ3B	AH4	GPIO_0_D14
GPIO_0_D0	V12	IO_3B/CLK1P/DIFFIO_RX_B23P			
GPIO_0_D2	W12	IO_3B/CLK1N/DIFFIO_RX_B23N			

Bank 4A

FPGA_CLK2_50	Y13	IO_4A/CLK2P/DIFFIO_RX_B31P
	AA13	IO_4A/CLK2N/DIFFIO_RX_B31N
GPIO_1_D0	Y15	IO_4A/CLK3P/DIFFIO_RX_B39P
GPIO_1_D2	AA15	IO_4A/CLK3N/DIFFIO_RX_B39N

Bank 8A

FPGA_CLK1_50 R159 0 FPGA_CLK3_50

	E11	IO_8A/CLK6P,FPLL_TL_FBP/DIFFIO_RX_T9P	IO_8A/FPLL_TL_CLKOUT0,FPLL_TL_CLKOUTP,FPLL_TL_FB/DIFFIO_TX_T4P	E8	
	D11	IO_8A/CLK6N,FPLL_TL_FBN/DIFFIO_RX_T9N	IO_8A/FPLL_TL_CLKOUT1,FPLL_TL_CLKOUTN/DIFFIO_TX_T4N	D8	
	D12	IO_8A/CLK7P/DIFFIO_RX_T1P			
	C12	IO_8A/CLK7N/DIFFIO_RX_T1N			

5CSEMA4U23C6N

Factory Default Configuration:
50MHz x2
25MHz x3
24MHz x2

Clock Generator

5 HPS_CLK1_25

5 HPS_CLK2_25

15 CLK_ENET_25

10 CLK_UB2_24

14 CLK_USB_24

CAD Note: Place near pin 3 and 5 (C3 & C322)

CAD Note: Place near IC power pin

VCC1P8 — L4 220 ohm, 0.3A — C247 4.7u 6.3V — C49 0.01u 50V — C50 1n 50V

L17 220 ohm, 0.3A — VCC3P3

C253 0.1u 10V — C244 0.1u 10V — C246 0.1u 10V — C252 4.7u 6.3V

C236 4p 6.3V DNI — C245 4p 6.3V DNI — Y1 24.00MHz

U15 CDCE937PWR

1	Xin/Clk	Y1	17	R47 18.2	CLK_UB2_24
20	Xout	Y2	15	R48 18.2	CLK_ENET_25
2	S0	Y3	14	R49 18.2	CLK_USB_24
CLK_I2C_SDA 19	S1/SDA	Y4	7	R54 18.2	HPS_CLK1_25
CLK_I2C_SCL 18	S2/SCL	Y5	8	R55 18.2	HPS_CLK2_25
4	Vctrl	Y6	12	R56 18.2	FPGA_CLK1_50
		Y7	11	R57 18.2	FPGA_CLK2_50

VDD 3 — Vddout 6, 10, 13 — GND 5, 9, 16

VCC3P3 — R174 4.7K — R175 4.7K

Default : I2C Address 0xDA/0xDB

Default: 24MHz
Default: 25MHz
Default: 24MHz
Default: 25MHz
Default: 25MHz
Default: 50MHz
Default: 50MHz

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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	FPGA Clock and Clock Generator	A2

Date: Monday, September 04, 2017 Sheet 6 of 23

FPGA JTAG INTERFACE

11 FPGA_TCK
11 FPGA_TMS
11 FPGA_TDI
11 FPGA_TDO

USB Blaster

10 FPGA_NCONFIG
10 FPGA_CONF_DONE

GPIO

3,6,19 GPIO_0_D[35..0]

ADC

18 ADC_CONVST
18 ADC_SCK
18 ADC_SDO
18 ADC_SDI

R171 0 DNI FPGA_TCK
C35 33p 50V DNI
R169 1K
VCC3P3
R176 10K DNI
R178 0 DNI
R177 1K DNI
C237 12p 50V DNI

Design Note:
Optional termination resistor for DCLK

CAD Note:
Place near FPGA DCLK pin

U1A

CYCLONE V SoC Configuration

Bank 3A

FPGA_TCK	AB5	TCK
FPGA_TMS	AC7	TMS
FPGA_TDI	W10	TDI
FPGA_TDO	Y9	TDO
EPCQ_DCLK	AA8	DCLK
EPCQ_AS_DATA0	AD7	AS_DATA0,ASDO/DATA0
EPCQ_AS_DATA1	AC6	AS_DATA1/DATA1
EPCQ_AS_DATA2	AC5	AS_DATA2/DATA2
EPCQ_AS_DATA3	AB6	AS_DATA3/DATA3
EPCQ_nCSO	AA6	NCSO/DATA4
GPIO_0_D7	Y4	IO_3A/DATA5/DIFFIO_TX_B2N
GPIO_0_D4	Y8	IO_3A/DATA6/DIFFIO_RX_B1N/DQ1B
GPIO_0_D8	Y5	IO_3A/DATA7/DIFFIO_TX_B2P/DQ1B
GPIO_0_D6	W8	IO_3A/DATA8/DIFFIO_RX_B1P/DQ1B
GPIO_0_D5	AB4	IO_3A/DATA9/DIFFIO_TX_B4N/DQ1B
GPIO_0_D10	T8	IO_3A/DATA10/DIFFIO_RX_B3N/DQSN1B
	AA4	IO_3A/DATA11/DIFFIO_TX_B4P
ADC_CONVST	U9	IO_3A/DATA12/DIFFIO_RX_B3P/DQS1B
ADC_SDO	AD4	IO_3A/DATA13/DIFFIO_TX_B6N/DQ1B
ADC_SCK	V10	IO_3A/DATA14/DIFFIO_RX_B5N/DQ1B
ADC_SDI	AC4	IO_3A/DATA15/DIFFIO_TX_B6P/DQ1B
U10		IO_3A/CLKUSR/DIFFIO_RX_B5P/DQ1B

Bank 5A

IO_5A/INIT_DONE/DIFFIO_RX_R2P	AA20
IO_5A/DEV_OE/DIFFIO_TX_R5P	AC24
IO_5A/DEV_CLRn/DIFFIO_TX_R5N/DQ1R	AB23
IO_5A/CRC_ERROR/DIFFIO_RX_R2N	Y19
IO_5A/nCEO/DIFFIO_TX_R3P/DQ1R	AE25

Bank 9A

CONF_DONE	J8	FPGA_CONF_DONE	R147 10K
nSTATUS	H8	FPGA_NSTATUS	R144 10K
nCONFIG	F7	FPGA_NCONFIG	R136 10K
nCE	E6		
MSEL0	J10	MSEL0	
MSEL1	H9	MSEL1	
MSEL2	G6	MSEL2	
MSEL3	K10	MSEL3	
MSEL4	K9	MSEL4	

5CSEMA4U23C6N

VCC3P3

VCC3P3
C40 0.1u 10V
U28
NC01 NC02 NC03 NC04 NC05 NC06 NC07 NC08
VCC GND
DATA0 DATA1 DATA2 DATA3 DCLK nCS
EPCS64
EPCQ_AS_DATA0 EPCQ_AS_DATA1 EPCQ_AS_DATA2 EPCQ_AS_DATA3 EPCQ_DCLK EPCQ_NCSO

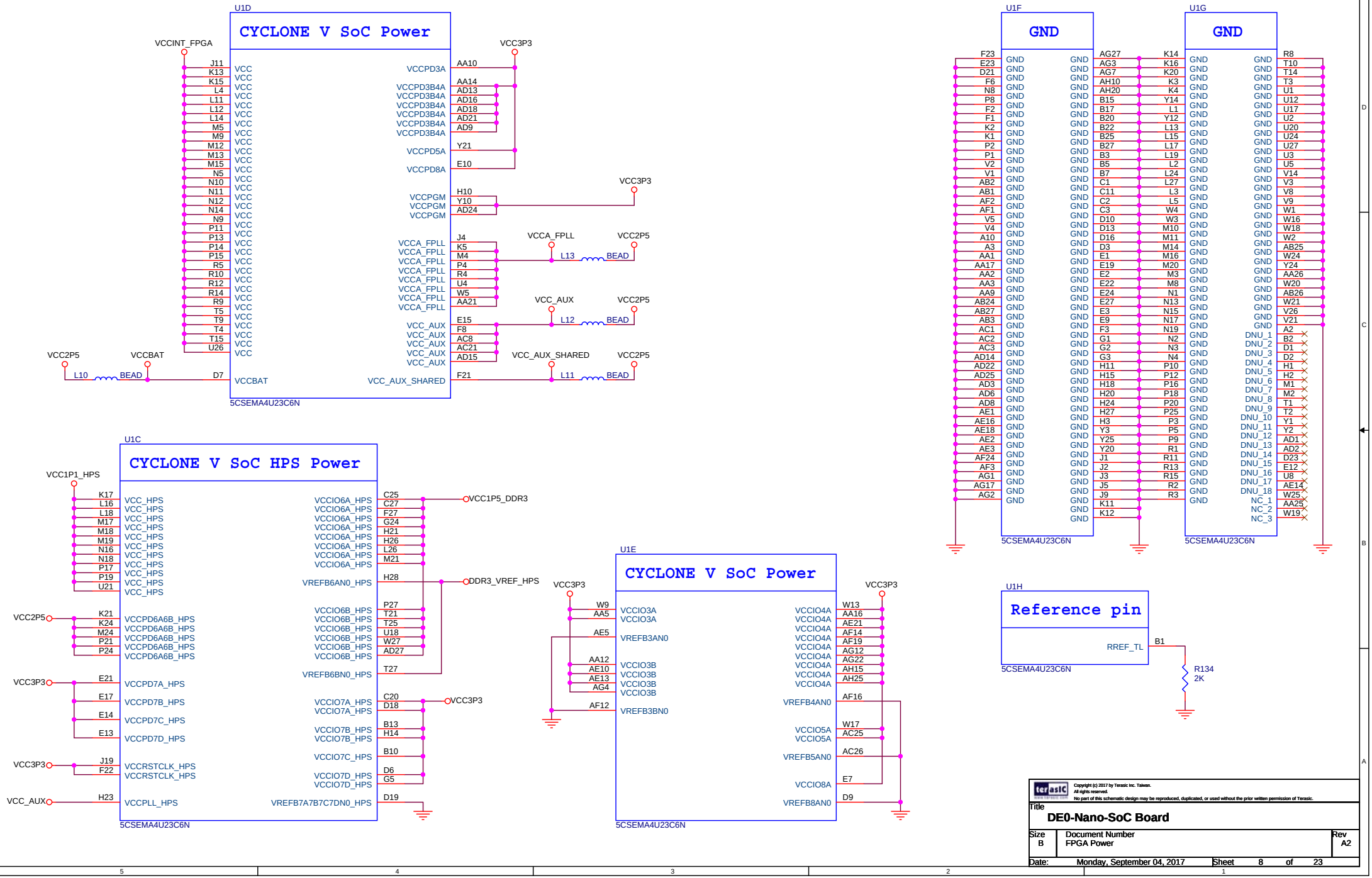
Default Setup MSEL[4:0] = 10010, AS Fast Mode

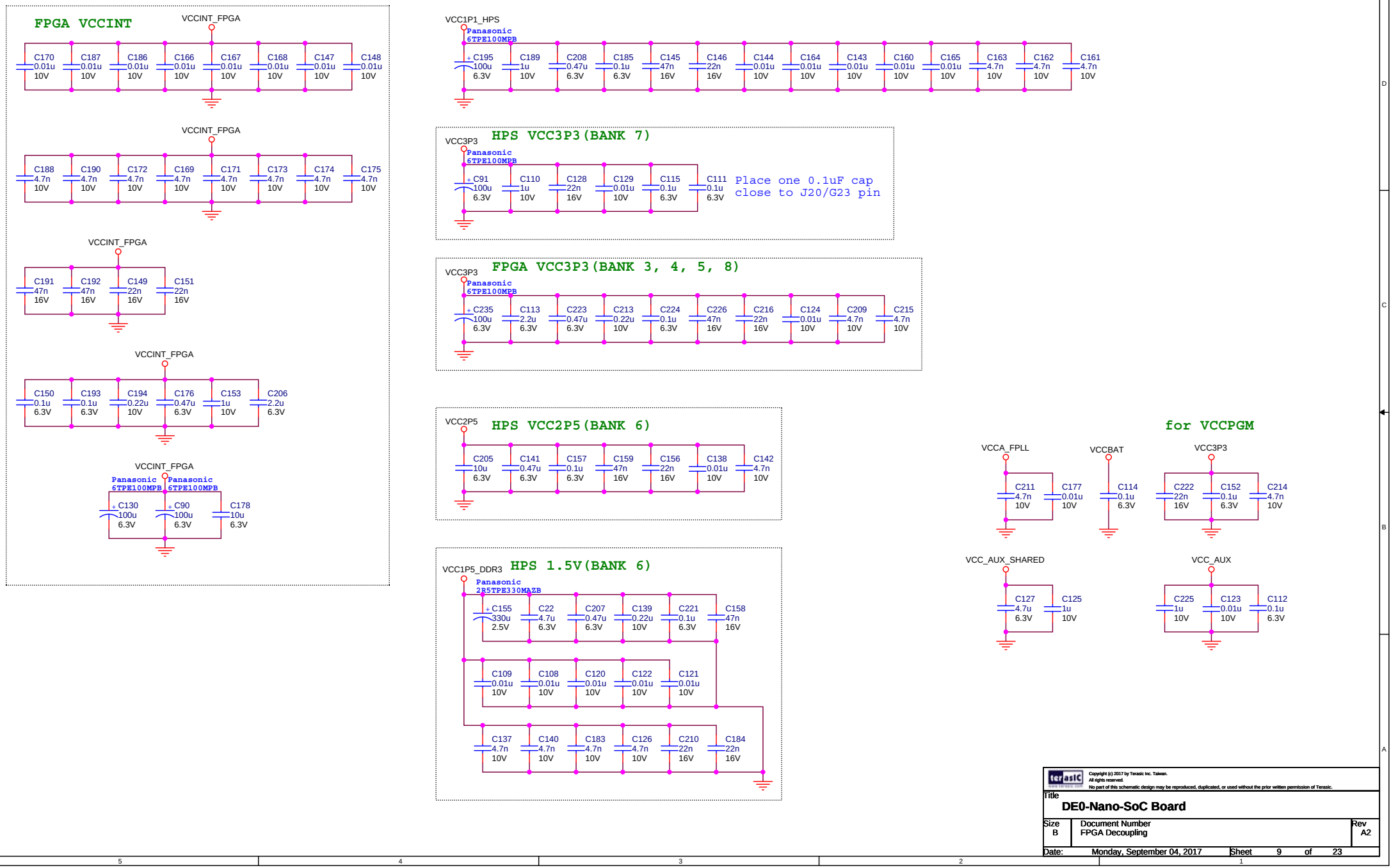
SW10 SW-DIP12
MSEL0 MSEL1 MSEL2 MSEL3 MSEL4 — R8 R9 R10 R11 R12 1K

R3 R4 R5 R6 R7 0 DNI MSEL0 MSEL1 MSEL2 MSEL3 MSEL4

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Title: DE0-Nano-SoC Board
Size: B
Document Number: FPGA Configuration and EPCS
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U1D
CYCLONE V SoC Power
VCCINT_FPGA
VCC3P3
VCC2P5
VCCBAT
L10
BEAD
D7
5CSEMA4U23C6N
U1C
CYCLONE V SoC HPS Power
VCC1P1_HPS
VCC1P5_DDR3
DDR3_VREF_HPS
VCC_AUX
5CSEMA4U23C6N
U1E
CYCLONE V SoC Power
5CSEMA4U23C6N
U1F
GND
5CSEMA4U23C6N
U1G
GND
5CSEMA4U23C6N
U1H
Reference pin
RREF_TL
B1
R134
2K
VCCA_FPLL
L13
VCC_AUX
L12
VCC_AUX_SHARED
L11
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Title
DE0-Nano-SoC Board
Size
B
Document Number
FPGA Power
Rev
A2
Date:
Monday, September 04, 2017
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8
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23

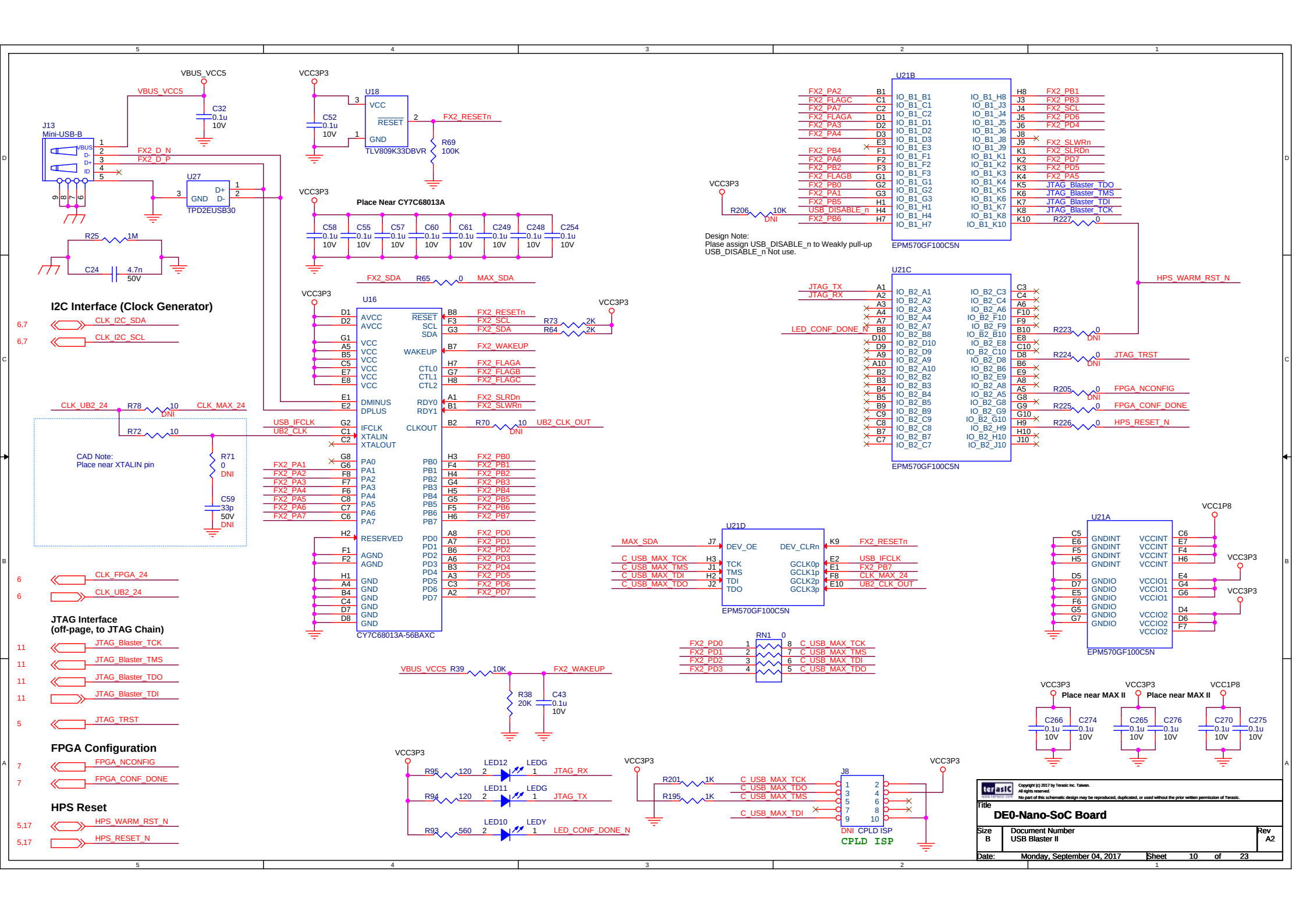


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Title
DE0-Nano-SoC Board

Size	Document Number	Rev
B	FPGA Decoupling	A2

Date: Monday, September 04, 2017 Sheet 9 of 23


VBUS_VCC5
C32 0.1u 10V
J13 Mini-USB-B
FX2_D_N
FX2_D_P
U27 TPD2EUSB30
R25 1M
C24 4.7n 50V
VCC3P3
U18 TLV809K33DBVR
C52 0.1u 10V
FX2_RESETn
R69 100K
Place Near CY7C68013A
C58 C55 C57 C60 C61 C249 C248 C254 0.1u 10V
FX2_SDA R65 0 MAX_SDA
U16 CY7C68013A-56BAXC
R73 2K
R64 2K
FX2_WAKEUP
FX2_FLAGA
FX2_FLAGB
FX2_FLAGC
FX2_SLRDn
FX2_SLWRn
CLK_UB2_24 R78 10 DNI CLK_MAX_24
R72 10
R71 0 DNI
C59 33p 50V DNI
CAD Note:
Place near XTALIN pin
USB_IFCLK
UB2_CLK
R70 10 DNI UB2_CLK_OUT
FX2_PA1
FX2_PA2
FX2_PA3
FX2_PA4
FX2_PA5
FX2_PA6
FX2_PA7
FX2_PB0
FX2_PB1
FX2_PB2
FX2_PB3
FX2_PB4
FX2_PB5
FX2_PB6
FX2_PB7
FX2_PD0
FX2_PD1
FX2_PD2
FX2_PD3
FX2_PD4
FX2_PD5
FX2_PD6
FX2_PD7
I2C Interface (Clock Generator)
6,7 CLK_I2C_SDA
6,7 CLK_I2C_SCL
6 CLK_FPGA_24
6 CLK_UB2_24
JTAG Interface
(off-page, to JTAG Chain)
11 JTAG_Blaster_TCK
11 JTAG_Blaster_TMS
11 JTAG_Blaster_TDO
11 JTAG_Blaster_TDI
5 JTAG_TRST
FPGA Configuration
7 FPGA_NCONFIG
7 FPGA_CONF_DONE
HPS Reset
5,17 HPS_WARM_RST_N
5,17 HPS_RESET_N
U21B EPM570GF100C5N
VCC3P3
R206 10K DNI
USB_DISABLE_n
Design Note:
Plase assign USB_DISABLE_n to Weakly pull-up
USB_DISABLE_n Not use.
R227 0
HPS_WARM_RST_N
U21C EPM570GF100C5N
JTAG_TX
JTAG_RX
LED_CONF_DONE_N
R223 0 DNI
R224 0 DNI JTAG_TRST
R205 0 DNI FPGA_NCONFIG
R225 0 FPGA_CONF_DONE
R226 0 HPS_RESET_N
U21D EPM570GF100C5N
MAX_SDA
C_USB_MAX_TCK
C_USB_MAX_TMS
C_USB_MAX_TDI
C_USB_MAX_TDO
USB_IFCLK
FX2_PB7
CLK_MAX_24
UB2_CLK_OUT
RN1 0
C_USB_MAX_TCK
C_USB_MAX_TMS
C_USB_MAX_TDI
C_USB_MAX_TDO
VBUS_VCC5 R39 10K FX2_WAKEUP
R38 20K C43 0.1u 10V
LED12 LEDG R95 120 JTAG_RX
LED11 LEDG R94 120 JTAG_TX
LED10 LEDY R93 560 LED_CONF_DONE_N
R201 1K
R195 1K
J8
DNI CPLD ISP
CPLD ISP
U21A EPM570GF100C5N
VCC1P8
VCC3P3
Place near MAX II
C266 C274 C265 C276 C270 C275 0.1u 10V
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Title
DE0-Nano-SoC Board
Size B
Document Number
USB Blaster II
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USB Blaster

10 JTAG_Blaster_TCK
10 JTAG_Blaster_TMS
10 JTAG_Blaster_TDO
10 JTAG_Blaster_TDI

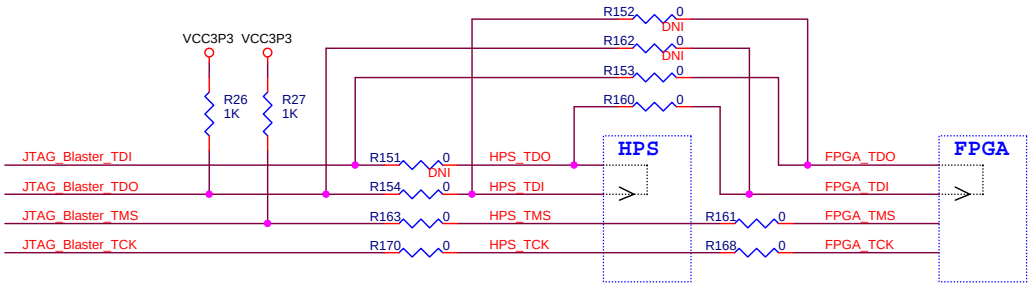
FPGA JTAG INTERFACE

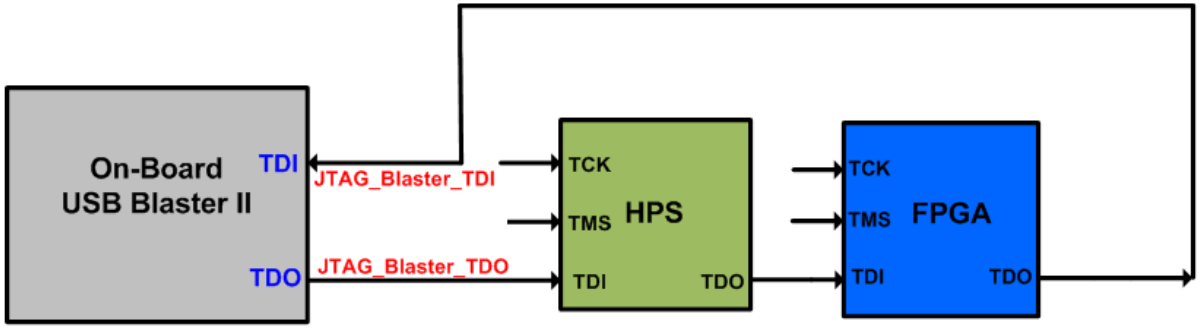
7 FPGA_TCK
7 FPGA_TMS
7 FPGA_TDI
7 FPGA_TDO

HPS JTAG INTERFACE

5 HPS_TCK
5 HPS_TMS
5 HPS_TDI
5 HPS_TDO

JTAG Chain





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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	JTAG Chain	A2

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DDR3 Interface (HPS)

HPS_DDR3_ADDR[14..0]
HPS_DDR3_DQ[31..0]
HPS_DDR3_BA[2..0]
HPS_DDR3_DM[3..0]
HPS_DDR3_DQS_P[3..0]
HPS_DDR3_DQS_N[3..0]
HPS_DDR3_CK_P
HPS_DDR3_CK_N
HPS_DDR3_CKE
HPS_DDR3_CS_N
HPS_DDR3_RESET_N
HPS_DDR3_WE_N
HPS_DDR3_RAS_N
HPS_DDR3_CAS_N
HPS_DDR3_ODT
HPS_DDR3_CS1_N
HPS_DDR3_CKE1
HPS_DDR3_ODT1

Note :
you can only swap the DQ signals
within x8 group (e.g. 0-7,8-15,16-23,24-31)
on the DDR3 chips

Note : you can swap the signals on the OCT resistor array
(include NC pin)

VCC1P5_DDR3 R22 2K HPS_DDR3_RESET_N V30
V1 HPS_DDR3_CK_P R132 100 HPS_DDR3_CK_N V2

CAD Note:
Place close to DDR3 chip

DDR3_VTT_HPS CN3 0.1u
DDR3_VTT_HPS CN2 0.1u
DDR3_VTT_HPS CN1 0.1u

DDR3_VTT_HPS RN5
9 HPS_DDR3_ADDR11 8 V11
10 HPS_DDR3_ADDR14 7 V13
11 HPS_DDR3_CAS_N 6 V16
12 HPS_DDR3_WE_N 5 V18
13 HPS_DDR3_ADDR0 4 V23
14 HPS_DDR3_ADDR9 3 V27
15 HPS_DDR3_ADDR13 2 V29
16 HPS_DDR3_ADDR2 1 V25
51

DDR3_VTT_HPS RN4
9 HPS_DDR3_ADDR10 8 V4
10 HPS_DDR3_ADDR12 7 V7
11 HPS_DDR3_BA1 6 V6
12 HPS_DDR3_ADDR1 5 V9
13 HPS_DDR3_ADDR4 4 V8
14 HPS_DDR3_ADDR6 3 V10
15 HPS_DDR3_RAS_N 2 V14
16 HPS_DDR3_ADDR8 1 V12
51

DDR3_VTT_HPS RN6
9 HPS_DDR3_ADDR7 8 V28
10 HPS_DDR3_ADDR5 7 V26
11 HPS_DDR3_ADDR3 6 V24
12 HPS_DDR3_BA2 5 V21
13 HPS_DDR3_BA0 4 V22
14 HPS_DDR3_CS_N 3 V19
15 HPS_DDR3_ODT 2 V17
16 HPS_DDR3_ODT1 1 V15
51
R21 51 HPS_DDR3_CS1_N V20
DNI

R146 4.7K HPS_DDR3_CKE1 V5
DNI
R149 4.7K HPS_DDR3_CKE V3

VCC1P5_DDR3 U11
DDR3_VREF_HPS
IS43TR16256A-15HBL

VCC1P5_DDR3 U6
DDR3_VREF_HPS
IS43TR16256A-15HBL

R150 240 DNI
R28 240
R114 240 DNI
R115 240

VCC1P5_DDR3
C204 2.2n 50V
C203 2.2n 50V
C201 2.2n 50V
C181 2.2n 50V
C199 2.2n 50V
C200 2.2n 50V
C198 3.3n 50V
C202 4.7n 50V
C182 0.01u 50V
C233 0.01u 50V
C220 0.01u 50V

VCC1P5_DDR3
C105 2.2n 50V
C104 2.2n 50V
C103 2.2n 50V
C101 2.2n 50V
C102 2.2n 50V
C100 2.2n 50V
C99 3.3n 50V
C97 4.7n 50V
C135 0.01u 50V
C134 0.01u 50V
C133 0.01u 50V

VCC1P5_DDR3
C219 0.1u 10V
C232 0.1u 10V
C231 0.1u 10V
C218 0.47u 10V
C228 0.47u 10V
C229 0.47u 10V

DDR3_VREF_HPS
C230 0.1u 10V
C197 0.1u 10V

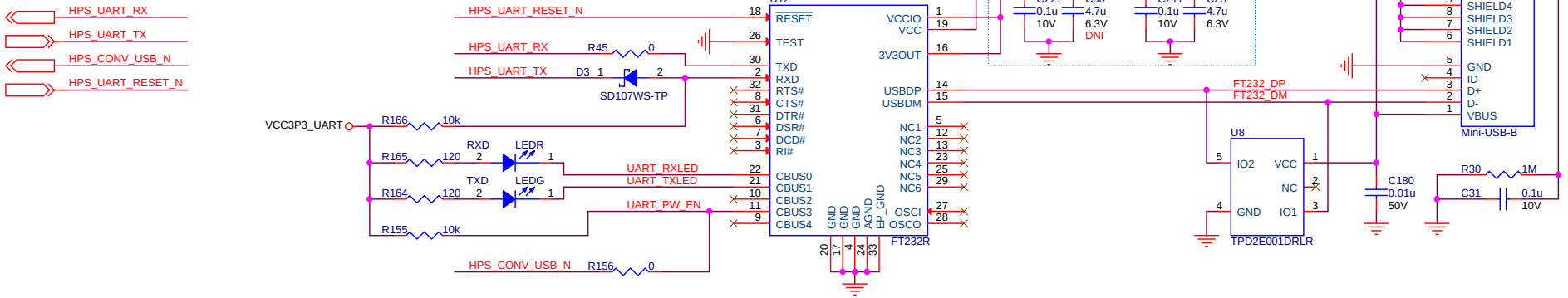
VCC1P5_DDR3
C117 0.1u 10V
C154 0.1u 10V
C131 0.1u 10V
C119 0.47u 10V
C118 0.47u 10V
C87 0.47u 10V

DDR3_VREF_HPS
C98 0.1u 10V
C132 0.1u 10V

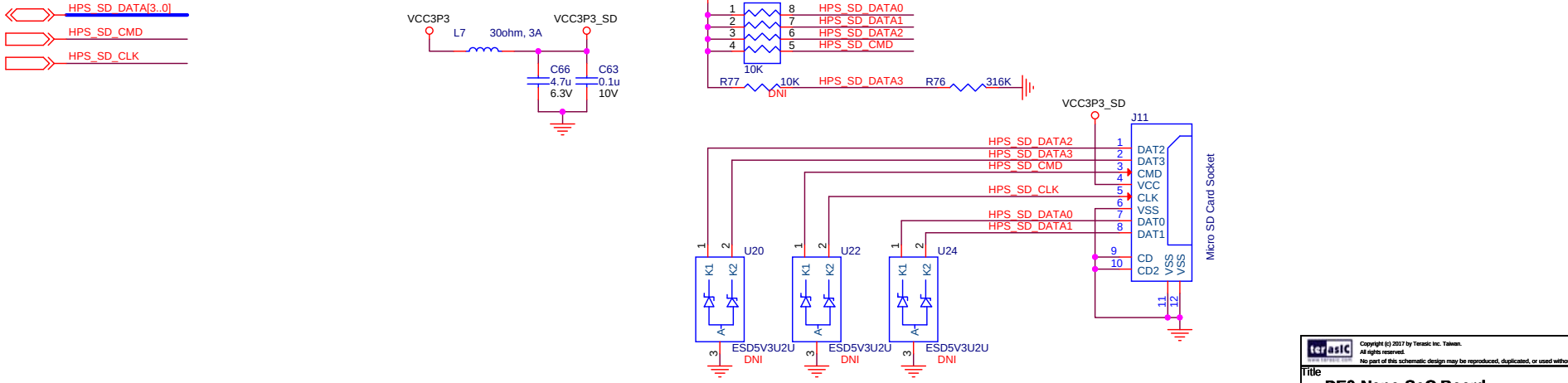
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Title: DE0-Nano-SoC Board
Size: B
Document Number: HPS : DDR3 SDRAM
Rev: A2
Date: Monday, September 04, 2017
Sheet 12 of 23

UART Interface

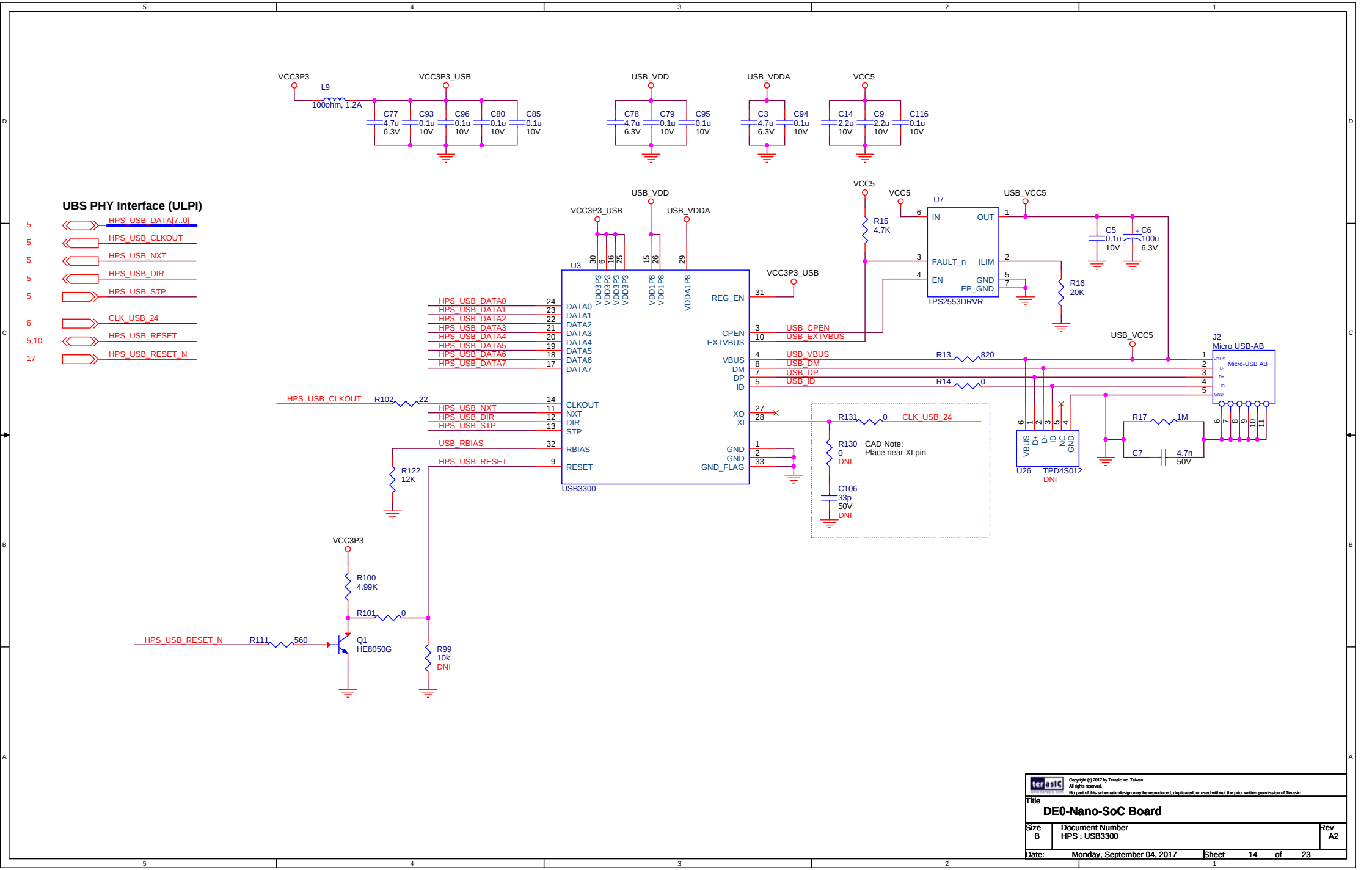


SD Card Interface



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Title	DE0-Nano-SoC Board	
Size B	Document Number HPS : UART to USB & SD CARD	Rev A2
Date: Monday, September 04, 2017	Sheet 13 of 23	



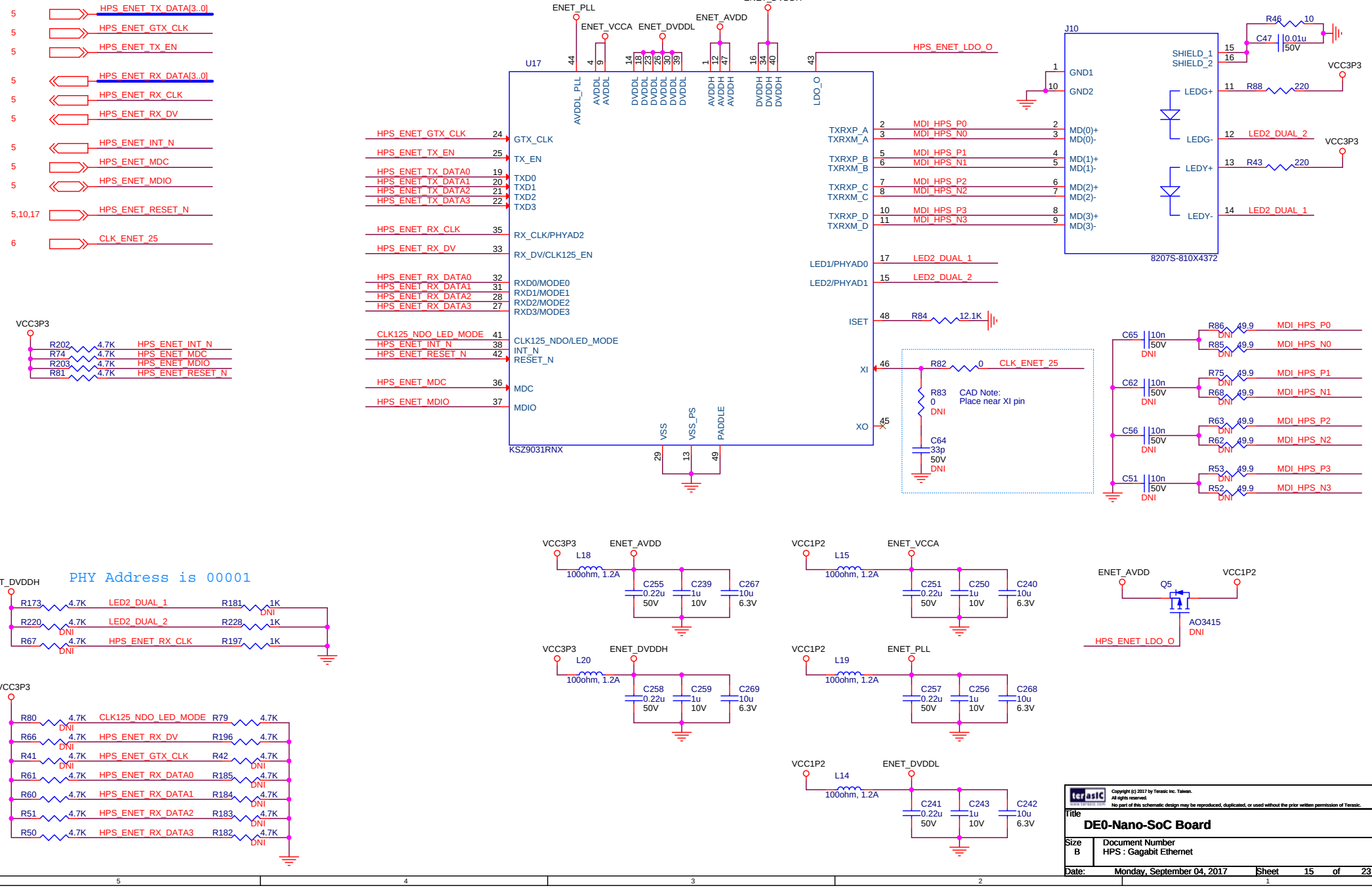
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Title
DE0-Nano-SoC Board

Size	Document Number	Rev
B	HPS : USB3300	A2

Date: Monday, September 04, 2017 Sheet 14 of 23

Ethernet PHY Interface (RGMII)



PHY Address is 00001

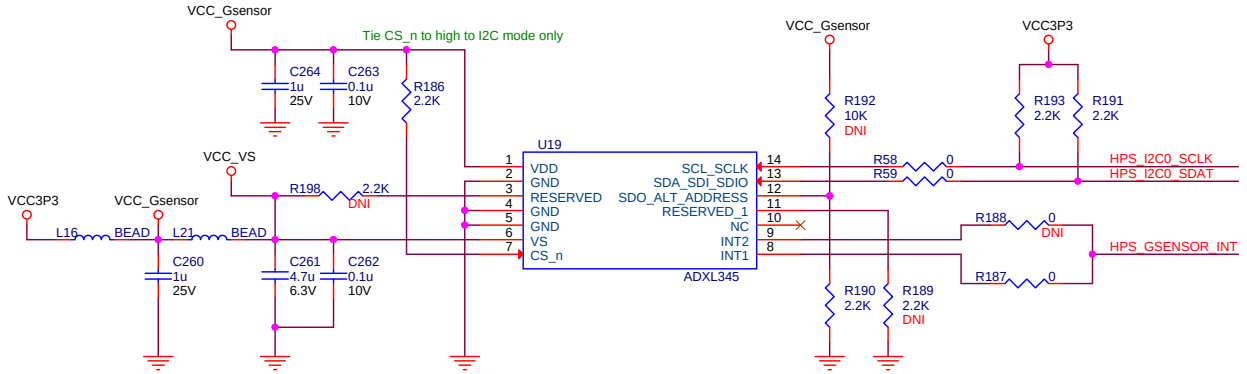
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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	HPS : Gagabit Ethernet	A2

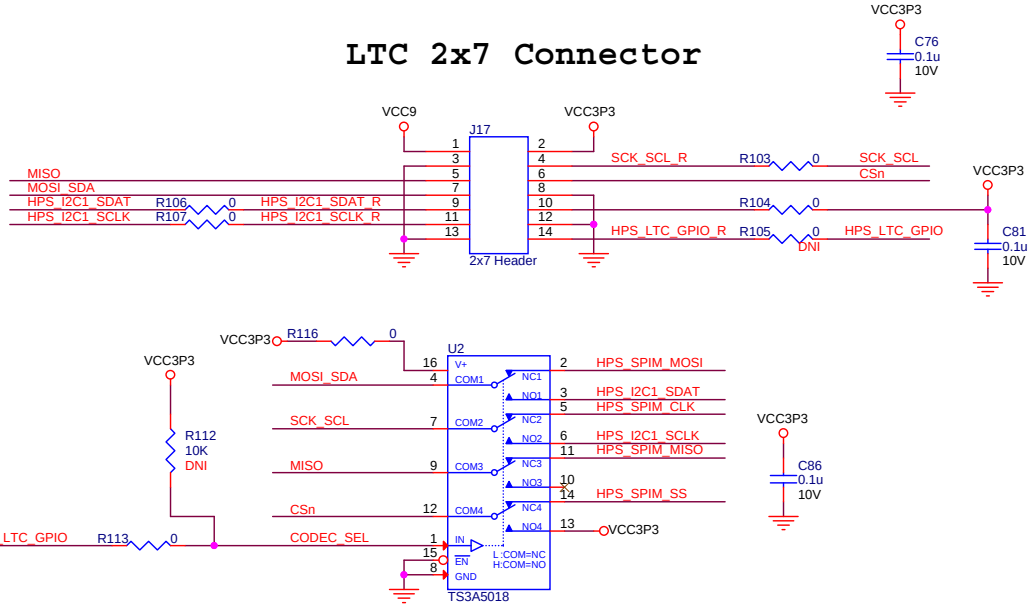
Date: Monday, September 04, 2017 Sheet 15 of 23

Digital Accelerometer



Default : I2C Address 0xA6/0xA7

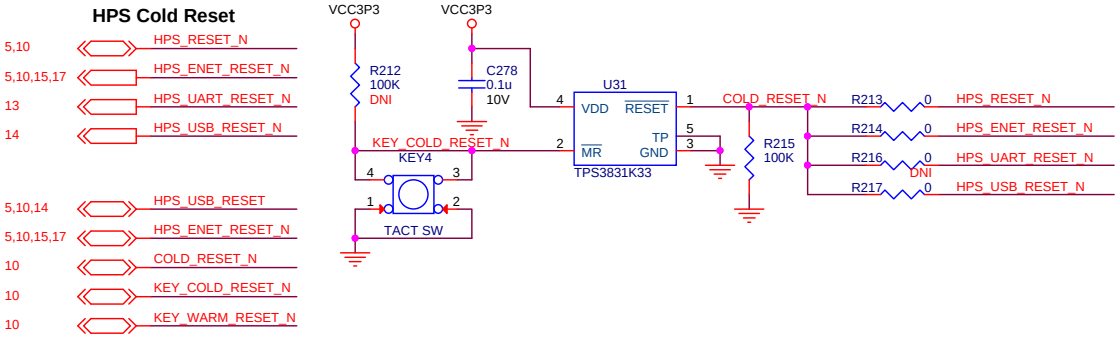
LTC 2x7 Connector



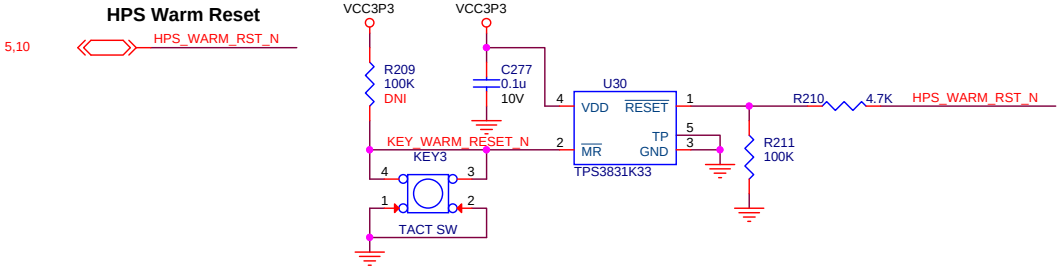
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Title: DE0-Nano-SoC Board		
Size: B	Document Number: HPS : Accelerometer, LTC Connector	Rev: A2
Date: Monday, September 04, 2017	Sheet 16 of 23	

HPS Cold Reset



HPS Warm Reset

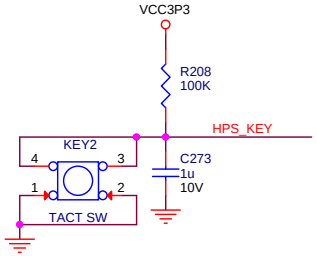


HPS Key and LED

5 HPS_KEY

5 HPS_LED

HPS User Button



HPS User LED

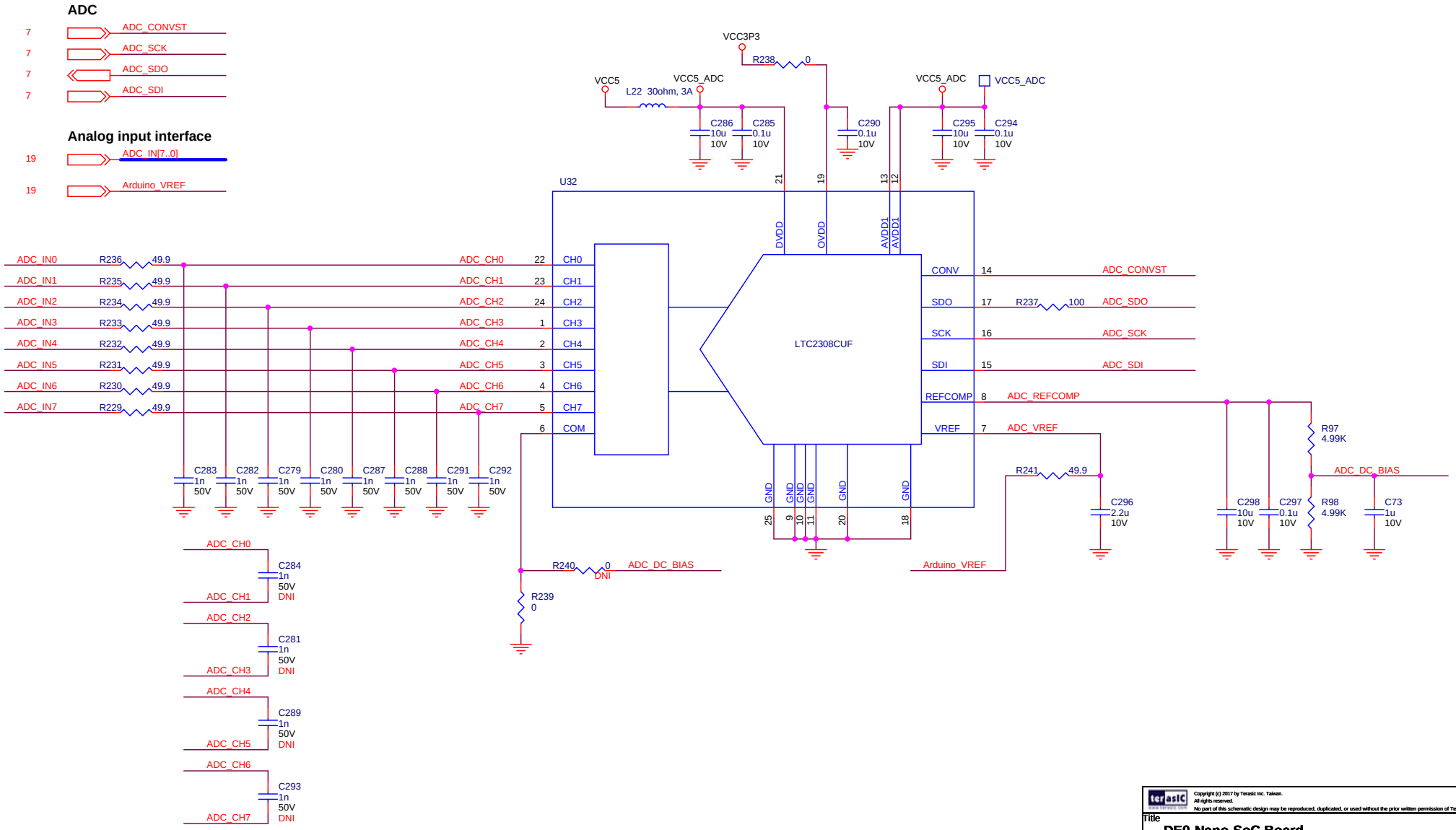


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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	HPS : BUTTON and LED	A2

Date: Monday, September 04, 2017 Sheet 17 of 23



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Title
DE0-Nano-SoC Board

Size	Document Number	Rev
B	FPGA : ADC1 - LTC2308 for ADC Header Analog input	A2

Date:	Monday, September 04, 2017	Sheet	18	of	23

GPIO

3,6,7 — GPIO_0_D[35..0]

3,6 — GPIO_1_D[35..0]

Arduino Digital Interface

3 — Arduino_IO[15..0]

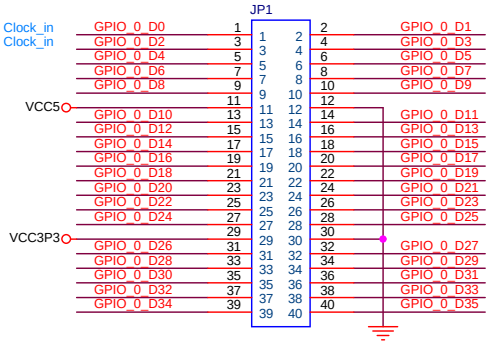
3 — Arduino_Reset_n

Analog input interface

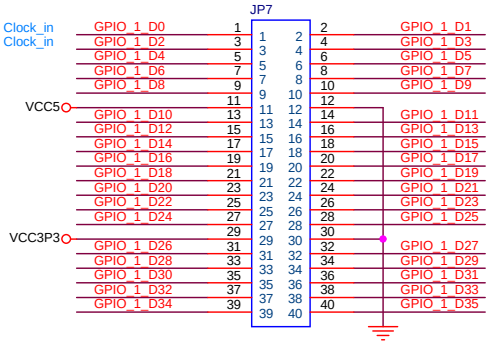
18 — ADC_IN[7..0]

18 — Arduino_VREF

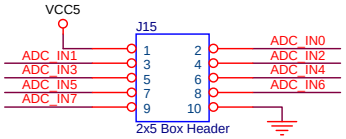
GPIO 0 Header

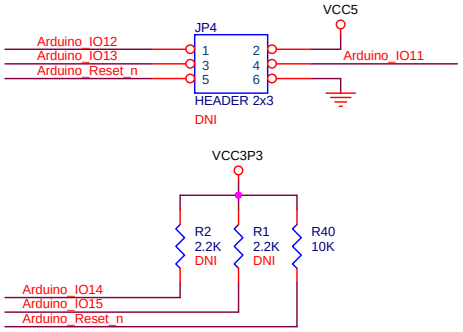


GPIO 1 Header

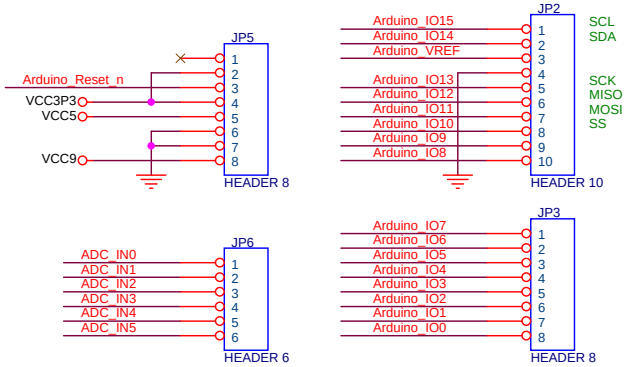


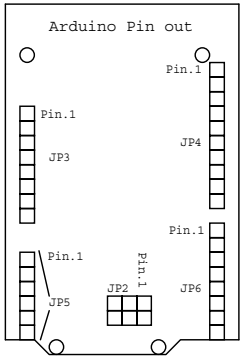
ADC Header





Arduino UNO Rev3





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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	FPGA : GPIO, ADC & Arduino Expansion Header	A2

Date: Monday, September 04, 2017 Sheet 19 of 23

KEY

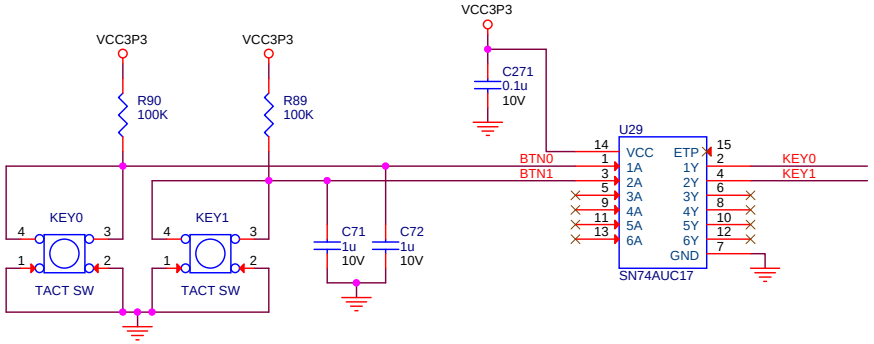
3 KEY[1..0]

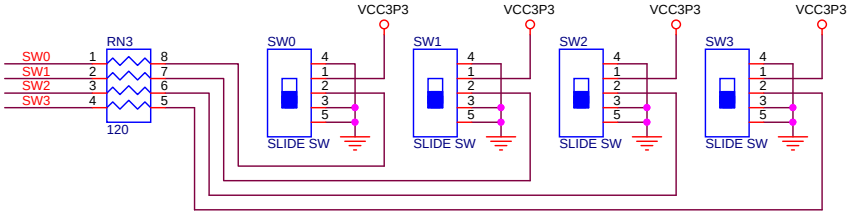
SWITCH

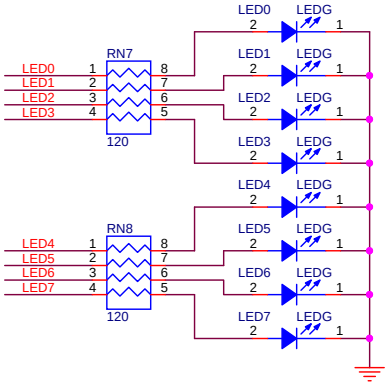
3 SW[3..0]

LED

3 LED[7..0]





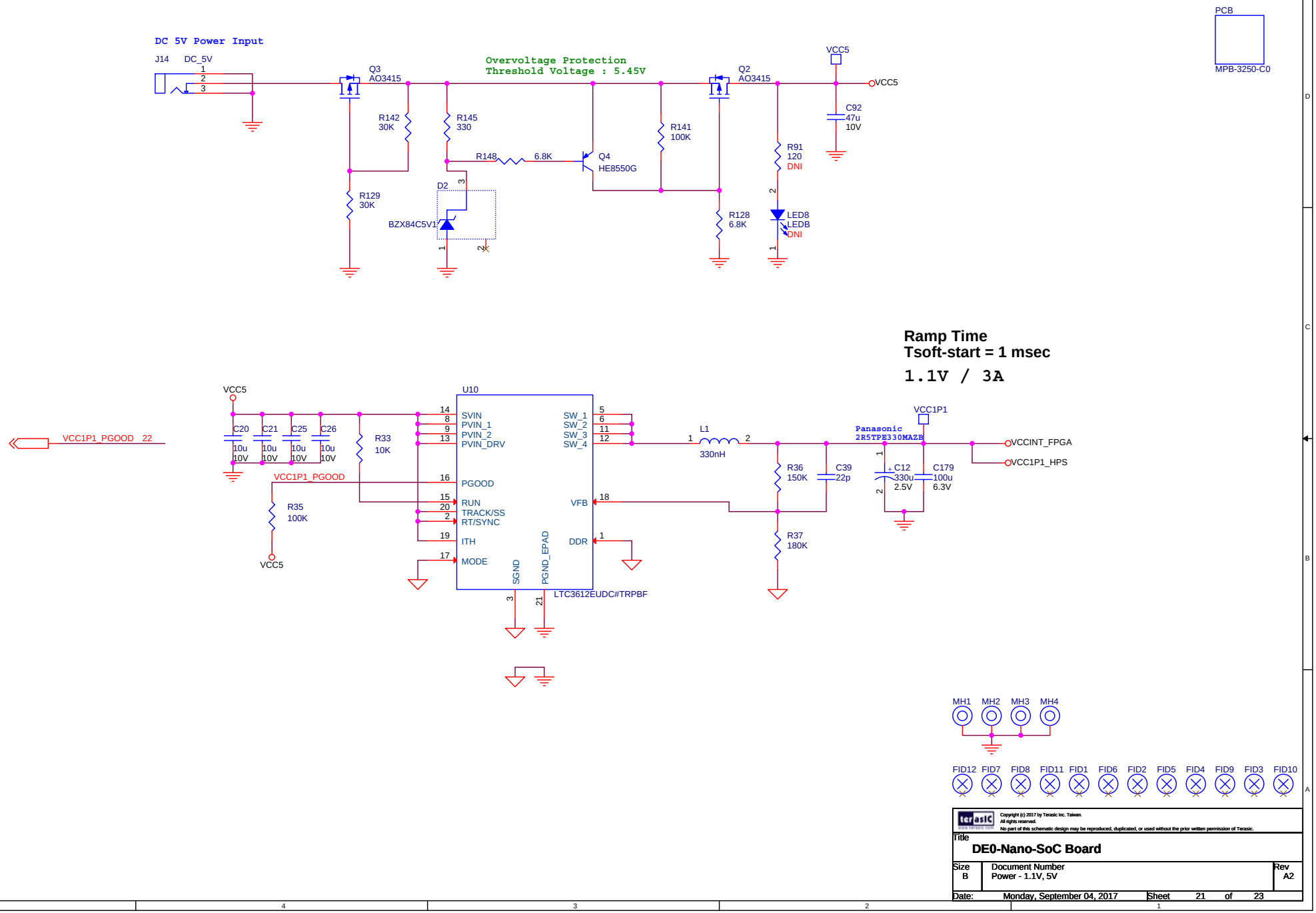


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Title: **DE0-Nano-SoC Board**

Size	Document Number	Rev
B	FPGA : LED, KEY, SW	A2

Date: Monday, September 04, 2017 Sheet 20 of 23



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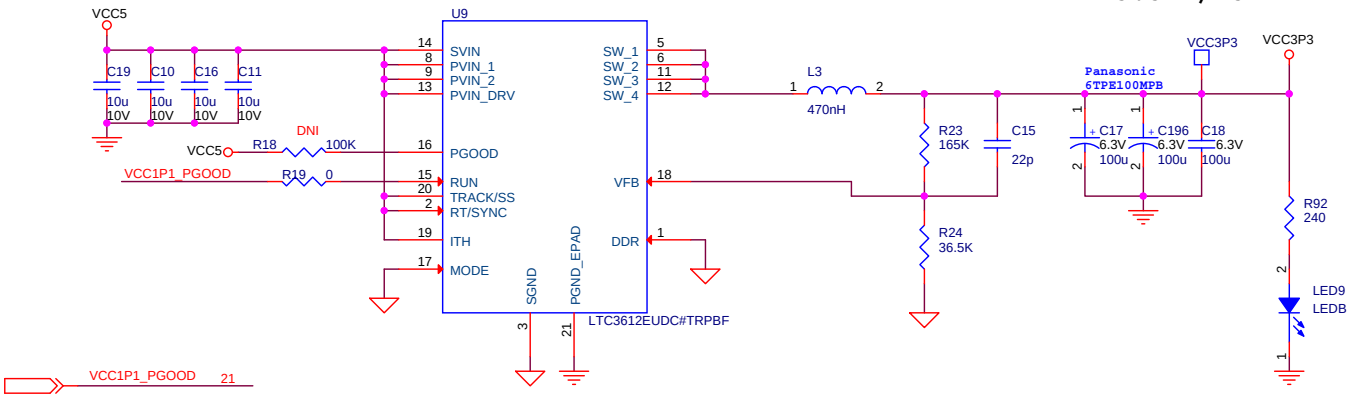
Title
DE0-Nano-SoC Board

Size	Document Number	Rev
B	Power - 1.1V, 5V	A2

Date: Monday, September 04, 2017 Sheet 21 of 23

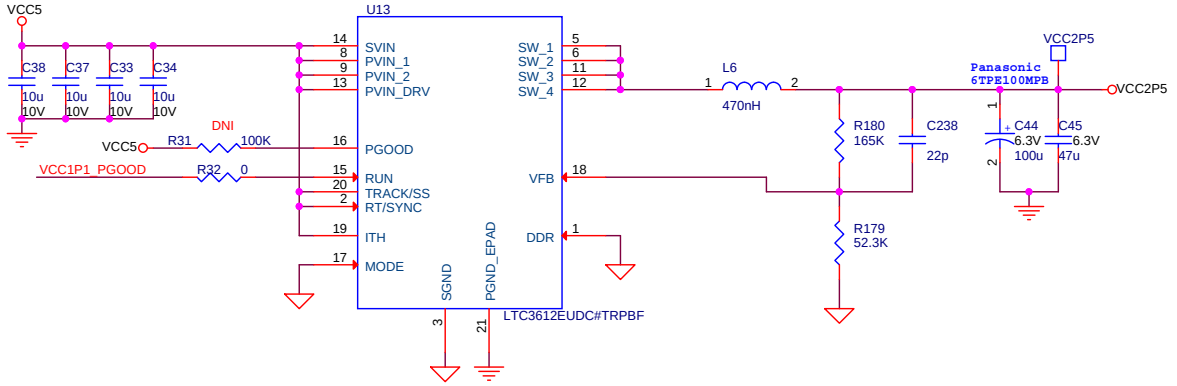
Ramp Time
Tsoft-start = 1 msec

3.3V / 3A



Ramp Time
Tsoft-start = 1 msec

2.5V / 3A



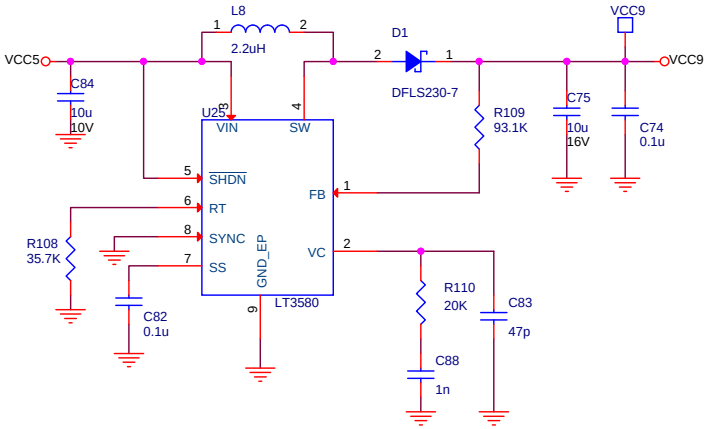
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Title
DE0-Nano-SoC Board

Size	Document Number	Rev
B	Power - 2.5V, 3.3V	A2

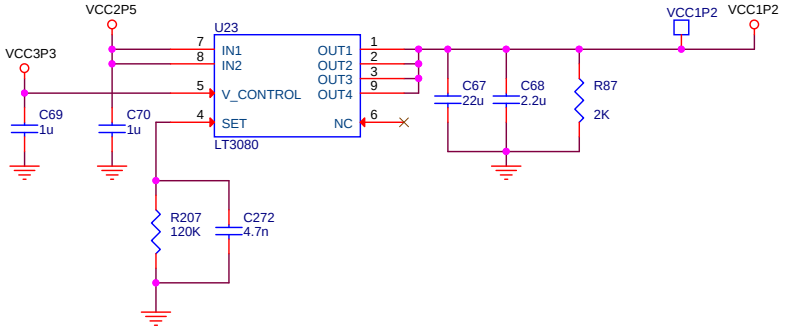
Date: Monday, September 04, 2017 Sheet 22 of 23

9V / 1A



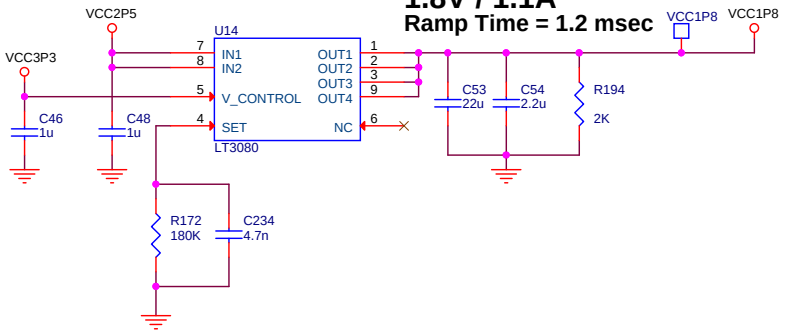
1.2V / 1.1A

Ramp Time = 0.8msec



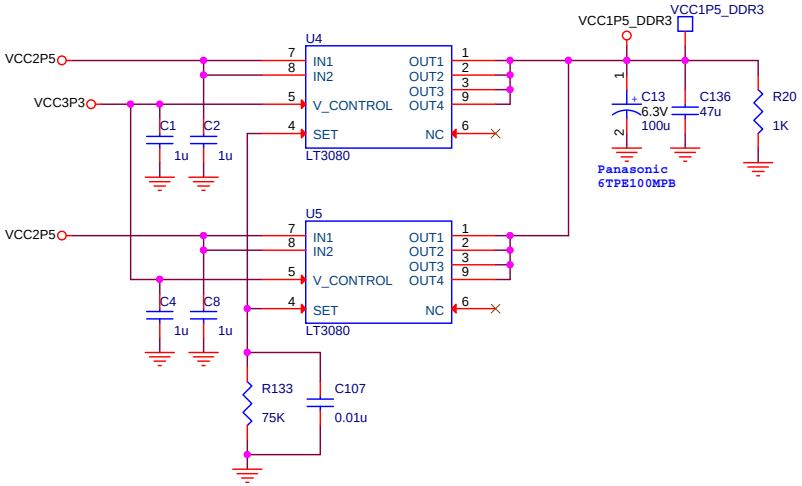
1.8V / 1.1A

Ramp Time = 1.2 msec

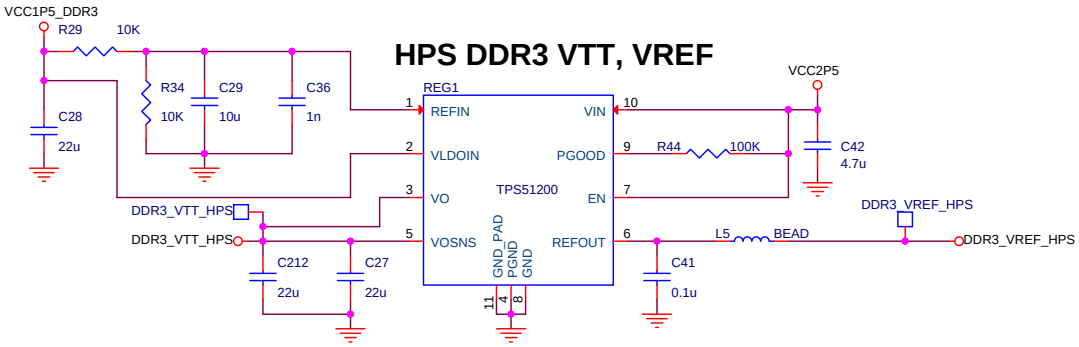


1.5V / 2.2A

Ramp Time = 1.27 msec



HPS DDR3 VTT, VREF



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Title
DE0-Nano-SoC Board

Size	Document Number	Rev
B	Power - 1.2V, 1.5V, 1.8V, 9V	A2

Date: Monday, September 04, 2017 Sheet 23 of 23